

1 Arithmetic determinant transport on the original tau complex

This calculation retains the arithmetic coefficient complex over the absolute pointed base $F_{1,\tau} = \{\tau, 1\}$. It computes the signed arithmetic/reference determinant term on the original finite theta source. The new formulas are a complete first and second variation, an exact four-window projection trace, and a quantitative bound in terms of the actual common source Gram. No arithmetic source multiplier is assumed bounded above or bounded away from zero.

1.1 Original objects and the fixed comparison

The retained tau-base construction has the four-point chart $+, - < \eta < \sigma$ and the coefficient sheaf $\mathcal{T}$ with $(\mathcal{T}_+, \mathcal{T}_-, \mathcal{T}_\eta, \mathcal{T}_\sigma) = (V, V, \mathcal{B}, 0)$. Its two nonzero restrictions are Θ and $J\Theta = \Theta\mathcal{F}$. The actual projective resolution gives

$$A_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi-\hat{\psi})} \mathcal{B}], \quad H^1 A_\tau(\mathcal{T}) = Q = \mathcal{B}/\Theta V. \quad (\text{AT1})$$

These are the fixed source objects of the original Tau Base note, equations (18)–(25), supplied in the source closure. The tensor resolution in its equations (46)–(48) retains the original differential signs and identifies its top cohomology with $Q^{\otimes k}$. The operation here uses that pushforward, with its arithmetic classes. Its separately computed restriction to σ remains an available comparison; it is not substituted for (AT1).

Fix an original actual packet h of full zero orders of $g = 2\xi$, and a positive integer k . Retain the full tensor-sum operator, its original cyclic annihilator $\chi = \chi_{h,k}$ of degree $q \geq 1$, and $E = \mathbb{C}[S]/(\chi)$ in the increasing-power remainder frame. The original construction supplies

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V}P = \eta_{h,k}J_N P, \quad \eta_{h,k}[P] = v_h^{\otimes k}P(A_h^{(k)}1), \quad (\text{AT2})$$

where $v_h = j_h(g/h)$ is the full invertible Taylor unit, J_N is literal remainder modulo χ , and $\eta_{h,k} : E \hookrightarrow E_h^{\otimes k}$ is the original injective full-jet observation, $E_h = \mathbb{C}[s]/(h)$. The original section $\sigma_h : E_h \hookrightarrow Q$ gives the separate arithmetic cohomology injection and equality

$$q^{(k)}\mathcal{V}P = \sigma_h^{\otimes k}\eta_{h,k}J_N P, \quad \sigma_h^{\otimes k}\eta_{h,k} : E \hookrightarrow Q^{\otimes k}. \quad (\text{AT2a})$$

The supplied arithmetic-input and cyclic-sum proofs construct these maps; all operations below keep them fixed. In particular the sum coordinate is $S = k/2 + iu$, with no division by k .

For $N \geq q - 1$, set $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{P}_{-1} = 0$, and let $B_N : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_N$ be multiplication by the literal monic χ . Write $s_N : E \rightarrow \mathcal{P}_N$ for the degree-less-than- q remainder section. Polynomial division proves

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0, \quad J_N s_N = I_E. \quad (\text{AT3})$$

The arithmetic norm has density $m_1(u) = m_{h,k}(u) = w_h^{*k}(u)$, $w_h(u) = |(g/h)(1/2 + iu)|^2/(2\pi)$. The retained Gamma reference has density

$$m_0(u) = r_{1/4,k}(u) = \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a). \quad (\text{AT4})$$

Its mass is $(2\pi)^{k/2}$; the arithmetic mass is the original μ_h^k . The source proofs included with this calculation establish finite moments of every degree and positivity almost everywhere for both densities. The calculation below only integrates polynomials against these two actual measures and their sum. It needs no estimate on the pointwise ratio m_1/m_0 .

For $0 \leq t \leq 1$ define, in the unchanged coefficient frame,

$$\begin{aligned} m_t &= (1-t)m_0 + tm_1, \quad \Delta m = m_1 - m_0, \quad v_N(u) = (1, S(u), \dots, S(u)^N), \\ M_N(t) &= \int v_N(u)^* v_N(u) m_t(u) du, \quad \Delta M_N = M_N(1) - M_N(0), \\ H_N(t) &= B_N^* M_N(t) B_N. \end{aligned} \quad (\text{AT5})$$

Every $M_N(t)$ is positive definite: a nonzero polynomial has finitely many real zeros, and m_t is positive almost everywhere. B_N is injective, so $H_N(t)$ is positive definite on its actual relation domain. When $N = q - 1$, that domain is zero; its determinant is one, its unique endomorphism is its inverse, and all compositions through that domain are zero. The real parameter t interpolates norms on (AT3); it changes neither the absolute base nor a cochain map.

The reference density in (AT4) is exactly the density denoted $r_{1/4}^{*k}$ in SP: with the retained convolution constant $c_a = 2^{1-2a}\Gamma(2a)$ one has

$$c_{1/4} = \sqrt{2\pi}, \quad r_{1/4}^{*k}(u) = \frac{c_{1/4}^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi} = m_0(u). \quad (\text{AT4a})$$

The original Gamma convolution identity therefore identifies all moments and the mass $(2\pi)^{k/2}$, without a mass adjustment. Writing $x = t$ and $y = u$, the precise coefficient arrow to the common-source notation of SP is $B_N^{\text{SP}} = I_N B_N^{\text{AT}}$, from $\mathcal{P}_{N-q}$ to $H = \mathcal{P}_{2q}$. Consequently

$$(B_N^{\text{SP}})^* M_* B_N^{\text{SP}} = B_N^* M_N B_N, \quad Q_N^{\text{SP}} = I_N B_N H_N^{-1} B_N^* I_N^* M_* = Q_N. \quad (\text{AT4b})$$

These identities follow from $M_N = I_N^* M_* I_N$. They include the zero relation domain, where all compositions are zero.

1.2 Canonical sections, determinant lines and full variations

Theorem 1.1. *On the fixed exact sequence (AT3), its least-norm section and quotient Gram are*

$$\begin{aligned} R_N(t) &= s_N - B_N H_N(t)^{-1} B_N^* M_N(t) s_N, \\ J_N R_N(t) &= I_E, \quad B_N^* M_N(t) R_N(t) = 0, \\ G_N(t) &= R_N(t)^* M_N(t) R_N(t) = (J_N M_N(t)^{-1} J_N^*)^{-1}, \\ V_N(t) &:= \det G_N(t) = \frac{\det M_N(t)}{\det H_N(t)}. \end{aligned} \quad (\text{AT6})$$

In the original polynomial frame the exact first and second variations are

$$\begin{aligned} \dot{R}_N &= -B_N H_N^{-1} B_N^* \Delta M_N R_N, \quad \dot{G}_N = R_N^* \Delta M_N R_N, \\ \ddot{G}_N &= -2R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N \preceq 0. \end{aligned} \quad (\text{AT7})$$

The exact logarithmic variation, with both trace terms retained, is

$$\begin{aligned} \frac{d}{dt} \log V_N &= \text{Tr}(M_N^{-1} \Delta M_N) - \text{Tr}(H_N^{-1} B_N^* \Delta M_N B_N) \\ &= \text{Tr}(G_N^{-1} R_N^* \Delta M_N R_N). \end{aligned} \quad (\text{AT8})$$

Its second derivative is

$$\begin{aligned} \frac{d^2}{dt^2} \log V_N &= -2 \text{Tr}(G_N^{-1} R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N) \\ &\quad - \text{Tr}(G_N^{-1} \dot{G}_N G_N^{-1} \dot{G}_N) \leq 0. \end{aligned} \quad (\text{AT9})$$

Proof. Every lift of $x \in E$ is $s_N x + B_N y$. Expanding its squared norm and completing its finite quadratic expression gives the unique minimizer $y = -H_N^{-1} B_N^* M_N s_N$. This proves the first two lines of (AT6). For the inverse expression, every polynomial vector p decomposes as $B_N y + R_N J_N p$. Orthogonality therefore gives $R_N^* M_N p = G_N J_N p$ for every p . Taking adjoints yields $M_N R_N = J_N^* G_N$. Substituting this into $J_N R_N = I_E$ gives $(J_N M_N^{-1} J_N^*) G_N = I_E$, which proves the inverse expression.

The determinant calculation preserves orientation. Put $r = N - q + 1$ and $L_N = [B_N \ s_N]$. In the ordered monomial bases, $[s_N \ B_N]$ has determinant one, since χ is monic and the lower diagonal block is triangular with diagonal one. Thus $\det L_N = (-1)^{qr}$. The upper-left block of $L_N^* M_N L_N$ is H_N , and its Schur complement is G_N . Consequently $\det H_N \det G_N = |(-1)^{qr}|^2 \det M_N = \det M_N$. Equivalently the map

$$\begin{aligned} \det \mathcal{P}_{N-q} \otimes \det E &\longrightarrow \det \mathcal{P}_N, \\ (y_1 \wedge \cdots \wedge y_r) \otimes (x_1 \wedge \cdots \wedge x_q) &\longmapsto B_N y_1 \wedge \cdots \wedge B_N y_r \wedge s_N x_1 \wedge \cdots \wedge s_N x_q \end{aligned} \quad (\text{AT10})$$

is the fixed determinant-line isomorphism. Replacing any lift by a relation does not change its wedge. Orthogonal lifts give the product metric just computed. This proves the determinant identity including the empty relation space, without changing any scalar mass.

Differentiate $J_N R_N = I$; then $\dot{R}_N$ lies in $\text{im } B_N$. Differentiating $B_N^* M_N R_N = 0$ determines its unique coefficient and proves the first identity of (AT7). On differentiating $R_N^* M_N R_N$, the two terms involving $\dot{R}_N$ vanish by that orthogonality. A second differentiation gives the two equal negative terms in (AT7), since ΔM_N is Hermitian. This proves the claimed second derivative and its sign. Applying the finite determinant derivative to (AT6) gives the first equality of (AT8); applying it to G_N gives the second. One more derivative gives (AT9). The first trace there is the squared Hilbert–Schmidt norm of $H_N^{-1/2} B_N^* \Delta M_N R_N G_N^{-1/2}$; the second is the squared Hilbert–Schmidt norm of $G_N^{-1/2} \dot{G}_N G_N^{-1/2}$. These identities establish their signs and retain the original forms. All inverse matrices are smooth on a neighborhood of the compact interval $[0, 1]$, by positivity and continuity. No infinite-dimensional inverse has been used. $\square$

The derivative $\dot{R}_N$ in (AT7) is a literal polynomial relation. Applying the original $\mathcal{V}$ in (AT2) therefore retains its original theta primitive and sends its arithmetic class to zero; $J^{(k)} \mathcal{V} R_N(t) = \eta_{h,k}$ for every t . At each nonempty original support label A , the cochain identities lift by $(A, P) \mapsto (A, fP)$, while $\tau \mapsto \tau$. Thus a relation-valued derivative has a supported zero class at label A . Its source vector and its full nonnegative energy in (AT7)–(AT9) remain present. These formulas compute their contribution rather than deleting it after taking cohomology.

1.3 The signed four-endpoint term on one retained source

Let $N_* = 2q$. Every space in (AT3) for the four endpoints sits in $\mathcal{P}_{N_*}$ by its actual coefficient inclusion. Give this common space the form $M_*(t) = M_{N_*}(t)$. Write $P_N(t)$ for its orthogonal projection onto $\mathcal{P}_N$, and $Q_N(t)$ for its orthogonal projection onto the relation subspace $\mathcal{R}_N = \chi \mathcal{P}_{N-q}$. The latter is zero at $N = q - 1$. For the coefficient inclusion $I_N : \mathcal{P}_N \hookrightarrow \mathcal{P}_{N_*}$, the formulas, with their domains retained, are

$$P_N = I_N M_N^{-1} I_N^* M_*, \quad Q_N = I_N B_N H_N^{-1} B_N^* I_N^* M_*, \quad C(t) = M_*(t)^{-1} \Delta M_*. \quad (\text{AT11})$$

Both P_N and Q_N are $M_*(t)$ -self-adjoint idempotents with their displayed images, verified by substitution. Since $\mathcal{R}_N \subset \mathcal{P}_N$, $P_N Q_N = Q_N P_N = Q_N$. Their difference is the orthogonal projection onto the finite canonical quotient section and has trace q . Cyclicity of finite matrix trace in (AT11) proves the exact common-source version of (AT8):

$$\frac{d}{dt} \log V_N(t) = \text{Tr}((P_N(t) - Q_N(t))C(t)). \quad (\text{AT12})$$

Put $i_0 = q - 1$, $i_1 = q$, $j_0 = 2q - 1$, and $j_1 = 2q$. Define the two positive operators

$$\begin{aligned} U(t) &= (Q_{j_0} - Q_{i_0}) + (Q_{j_1} - Q_{i_1}), \\ W(t) &= (P_{j_0} - P_{i_0}) + (P_{j_1} - P_{i_1}). \end{aligned} \quad (\text{AT13})$$

Each difference in this display is an orthogonal projection, because the two spaces in its pair are nested in the same Hilbert space. Each has rank q . The sums need not be projections and are not asserted to be orthogonal sums; they are positive operators with $\text{Tr } U = \text{Tr } W = 2q$.

Write

$$\begin{aligned}\mathcal{B}(t) &= \log \frac{V_{i_0}(t)V_{i_1}(t)}{V_{j_0}(t)V_{j_1}(t)}, \quad T_N = V_N(1)/V_N(0), \\ \Delta\mathcal{B} &= \mathcal{B}(1) - \mathcal{B}(0) = \log \frac{T_{q-1}T_q}{T_{2q-1}T_{2q}}.\end{aligned}\tag{AT14}$$

Subtracting the four exact formulas (AT12), with their original signs, and integrating gives

$$\boxed{\Delta\mathcal{B} = \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt.}\tag{AT15}$$

This is the actual arithmetic correction in the audit's formula (54). It compares the two added relation windows with the two added polynomial windows. Concavity of each individual $\log V_N(t)$ in (AT9) does not assign a sign to their signed combination in (AT15); that combination has now been computed.

There is also an exact pointwise integral, using no division by the arithmetic density. Define

$$\begin{aligned}k_N(t, u) &= v_N(u)M_N(t)^{-1}v_N(u)^*, \\ r_N(t, u) &= |\chi(S(u))|^2 v_{N-q}(u)H_N(t)^{-1}v_{N-q}(u)^*, \\ d_N(t, u) &= k_N(t, u) - r_N(t, u), \quad r_{q-1}(t, u) = 0, \\ d_N(t, u) &= v_N(u)R_N(t)G_N(t)^{-1}R_N(t)^*v_N(u)^* \geq 0.\end{aligned}\tag{AT16}$$

The last equality follows by writing the orthogonal splitting of M_N^{-1} as $B_N H_N^{-1} B_N^* + R_N G_N^{-1} R_N^*$; multiplying by M_N and evaluating on $\text{im } B_N \oplus \text{im } R_N$ proves that inverse identity. In particular $\int d_N(t, u)m_t(u) du = q$. Consequently (AT8) also reads

$$\frac{d}{dt} \log V_N(t) = \int d_N(t, u) \Delta m(u) du, \quad \Delta\mathcal{B} = \int_0^1 \int (d_{i_0} + d_{i_1} - d_{j_0} - d_{j_1})(t, u) \Delta m(u) du dt.\tag{AT17}$$

All coefficients of these fixed-degree polynomials in u and $\bar{S}(u)$ are bounded for $t \in [0, 1]$. Since $|\Delta m| \leq m_0 + m_1$ and these measures have the required finite moments, absolute integrability proves the exchange of integrals. This remains valid at every zero of m_1 .

1.4 The full spectral and actual-overlap bound for the correction

Let $n = 2q + 1$ and let $b_1 \leq \dots \leq b_n$ be all eigenvalues of $D_* = M_*(0)^{-1}M_*(1)$ in its original $M_*(0)$ inner product. This operator is positive because $v^* M_*(0) D_* v = v^* M_*(1) v > 0$ for $v \neq 0$. Retain $\kappa_* = b_n/b_1$ and put

$$\begin{aligned}c_j(t) &= \frac{b_j - 1}{1 - t + t b_j}, & d(t) &= c_n(t) - c_1(t), \\ \mathcal{L}_q(D_*) &= \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}, \\ S_{\text{AW}}(t) &= c_{q+2}(t) - c_q(t) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(t) - c_j(t)), \\ s(t) &= \dim(\text{ran } U(t) \cap \text{ran } W(t)), \\ S_{\text{SP}}(t) &= \frac{d(t)}{2} \min \left\{ 4q - 2s(t), \sqrt{(2q+1)(8q-4-2\text{Tr}(U(t)W(t)))} \right\}, \\ \mathcal{J}_{h,k} &= \int_0^1 \min\{S_{\text{AW}}(t), S_{\text{SP}}(t)\} dt.\end{aligned}\tag{AT18a}$$

An empty sum for $q = 1$ is zero. Every quantity uses the unchanged common source $H = \mathcal{P}_{2q}$ and the actual relation maps in (AT11). The complete spectra and $\mathcal{L}_q$ bound were proved in AW7 and AW13–AW16; the actual overlap terms were proved in SP8–SP10. Their full proofs are propagated here to the original endpoint estimate, so neither calculation is replaced by a reference.

The exact trace norm and the centered Hilbert–Schmidt estimate are also retained before taking a minimum. On this same source define

$$\begin{aligned} S_{\text{tr}}(t) &= \frac{d(t)}{2} \|U(t) - W(t)\|_1, \\ \Sigma(t) &= \text{Tr}(C(t)^2) - \frac{(\text{Tr } C(t))^2}{2q+1}, \\ S_{\text{HS}}(t) &= \sqrt{\Sigma(t) \text{Tr}((U(t) - W(t))^2)}. \end{aligned} \tag{AT18c}$$

The trace-norm inequality follows by subtracting the midpoint of the extreme eigenvalues from the metric derivative, since the trace of the signed projection difference is zero. In an orthonormal eigenbasis of that difference, every diagonal derivative entry is bounded in modulus by half the spectral diameter; summing with the absolute eigenvalues proves the assertion. For the other bound subtract the trace mean $\text{Tr}(C(t))I/(2q+1)$ instead. The square of its Hilbert–Schmidt norm in the unchanged source metric is exactly $\Sigma(t)$. Hilbert–Schmidt Cauchy–Schwarz with the signed difference proves that its trace pairing is bounded by $S_{\text{HS}}(t)$. Positivity of both factors under the square root follows from their self-adjointness. Thus both functions bound the same absolute signed derivative on general positive source forms; the angle term below still uses the specified moment measures. These are RMT10a–10b and ISM25–26 through the already proved original coefficient and projection maps.

The original-moment angle calculation supplies a further simultaneous bound at this same claim site. Put $a_q = 2q - 1$ and

$$\begin{aligned} A_{h,k}(t) &= a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}} d(t), \\ \mathcal{J}_{h,k}^{(5)} &= \int_0^1 \min\{S_{\text{AW}}(t), S_{\text{tr}}(t), S_{\text{SP}}(t), S_{\text{HS}}(t), A_{h,k}(t)\} dt, \\ \mathcal{I}_{h,k} &= \int_0^1 \min \left\{ d(t), \frac{S_{\text{AW}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}}, \frac{S_{\text{SP}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}}, \right. \\ &\quad \left. \frac{S_{\text{tr}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}}, \frac{S_{\text{HS}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}} \right\} dt. \end{aligned} \tag{AT18b}$$

These specialize RMT1–RMT18 through $a_{\text{RMT}} = a_q$, $x = t$, $C_x = C(t)$ and $(U_x, W_x) = (U(t), W(t))$. The density and relation arrows were proved in (AT4a)–(AT4b), so every source moment and original cohomology map agrees. This application uses the actual moment measures (AT4)–(AT5). It does not impose moment structure on an arbitrary positive coefficient Gram.

Theorem 1.2. *The exact correction on the original finite source satisfies*

$$|\Delta \mathcal{B}| \leq \mathcal{J}_{h,k}^{(5)} \leq \mathcal{J}_{h,k} \leq \mathcal{L}_q(D_*) \leq (2q - 1) \log \kappa_*. \tag{AT18}$$

Moreover $\mathcal{J}_{h,k} \leq \int_0^1 S_{\text{SP}}(t) dt$. The original Gamma and arithmetic endpoint quantities consequently obey

$$\max\{\mathcal{B}_{h,k}^\Gamma - \mathcal{J}_{h,k}^{(5)}, \mathcal{B}_{h,k}^\Gamma + j_{\ell_N} - e_{\ell_N}\} \leq \mathcal{B}_{h,k}^{\text{ar}} \leq \min\{\mathcal{B}_{h,k}^\Gamma + \mathcal{J}_{h,k}^{(5)}, \mathcal{B}_{h,k}^\Gamma + j_{\ell_N} + e_{\ell_N}\}. \tag{AT19}$$

Here the constructed j_{ℓ_N}, e_{ℓ_N} are defined and proved in (AT19ba)–(AT19bc) below.

Proof. The relation flag $\chi \mathcal{P}_0 \subset \chi \mathcal{P}_{q-1} \subset \chi \mathcal{P}_q \subset H$ has dimensions $1, q, q+1, 2q+1$. On its successive orthogonal pieces the operator $U = Q_{2q-1} + Q_{2q} - Q_q$ has values $1, 2, 1, 0$ with multiplicities

$1, q-1, 1, q$. On the orthogonal pieces of $\mathcal{P}_{q-1} \subset \mathcal{P}_q \subset \mathcal{P}_{2q-1} \subset H$, W has values $0, 1, 2, 1$ with multiplicities $q, 1, q-1, 1$. Nested orthogonal projections act as identity or zero on these pieces, which proves the assertions including $q=1$. In particular

$$\begin{aligned} \text{rank } U &= \text{rank } W = q+1, & \text{Tr } U &= \text{Tr } W = 2q, & \text{Tr } U^2 &= \text{Tr } W^2 = 4q-2, \\ A &= U - W, & \text{Tr } A &= 0, & \text{Tr } A^2 &= 8q-4-2\text{Tr}(UW). \end{aligned} \quad (\text{AT20a})$$

Thus the equal mass $2q$ is an operator trace, and its multiplicity-two subspace of dimension $q-1$ is retained in each rank calculation.

The identity $M_*C = \Delta M_* = \Delta M_*^*$ proves that C is self-adjoint in $M_*(t)$. The exact factorization $M_*(t) = M_*(0)((1-t)I + tD_*)$ gives $C = ((1-t)I + tD_*)^{-1}(D_* - I)$. A basis of $M_*(0)$ -orthonormal eigenvectors of D_* remains orthogonal in $M_*(t)$, with positive squared lengths $1-t+tb_j$, proving that the displayed $c_j(t)$ are its eigenvalues. Differentiation in b_j gives $(1-t+tb_j)^{-2} > 0$, so they retain their order.

For any $M_*(t)$ -orthogonal projection P of rank r , choose an orthonormal eigenbasis e_j of C . Its diagonal values $a_j = \langle e_j, Pe_j \rangle$ lie in $[0, 1]$ and sum to r . Thus $\text{Tr}(PC) = \sum_j c_j a_j$ lies between the sum of the smallest r and the sum of the largest r eigenvalues: moving a fixed amount of mass from a smaller coefficient to a larger coefficient increases the sum, until these two extrema are obtained. This also proves the assertion when there are ties. Either $Z = U$ or $Z = W$ is the sum of its range projection of rank $q+1$ and its eigenvalue-two projection of rank $q-1$. Applying the just-proved inequality gives

$$\sum_{j=1}^{q+1} c_j + \sum_{j=1}^{q-1} c_j \leq \text{Tr}(ZC) \leq \sum_{j=q+1}^{2q+1} c_j + \sum_{j=q+3}^{2q+1} c_j. \quad (\text{AT20b})$$

Subtracting the extrema for the two original operators, retaining both sums before their common middle term cancels, proves $|\text{Tr}(AC)| \leq S_{AW}(t)$.

Both ranges have dimension $q+1$ in dimension $2q+1$; hence $s(t) \geq 1$. Let E_L be the orthogonal projection onto their intersection. Each of U, W dominates its range projection, since its nonzero eigenvalues are $1, 2$; that projection dominates E_L . Therefore $U - E_L, W - E_L$ are positive, of trace $2q - s(t)$, and

$$\|A\|_1 \leq 4q - 2s(t), \quad \|A\|_1 \leq \sqrt{(2q+1)\text{Tr } A^2}. \quad (\text{AT20c})$$

The first inequality is the trace-norm triangle inequality applied to their difference. The second is Cauchy-Schwarz on all $2q+1$ real eigenvalues of A . Subtracting the midpoint $a = (c_1 + c_n)/2$ from C leaves $\text{Tr}(AC)$ unchanged by $\text{Tr } A = 0$. In an orthonormal eigenbasis f_j of A with eigenvalues α_j , $|\text{Tr}(A(C - aI))| \leq \sum_j |\alpha_j| |\langle f_j, (C - aI)f_j \rangle| \leq d(t)\|A\|_1/2$. Together with (AT20a)–(AT20c), this proves the full pointwise bound

$$|\text{Tr}((U - W)C)| \leq \min\{S_{AW}(t), S_{SP}(t)\}. \quad (\text{AT20})$$

No pairwise commutation of U, W, C has been used. The Gram inverses and projectors are smooth on the compact interval; their ranks and intersection dimensions are Borel, since ranks are specified by vanishing and nonvanishing minors. All terms are bounded, hence the displayed minimum is integrable. Integrating (AT20) in (AT15) gives $|\Delta\mathcal{B}| \leq \mathcal{J}_{h,k}$ for these two positive forms, even without moment structure. On the actual measures, RMT11 retains the two full q -angle lists of the crossed pairs $(q-1, 2q)$ and $(q, 2q-1)$, including the designated zero angle, and proves $|\mathcal{B}'| \leq A_{h,k}$. Its strict positivity proof applies here: if $\mathcal{B} = 0$, the first pair has identical minimum sections, putting 1 in the orthogonal complement of $\chi\mathcal{P}_q$. The polynomial $\chi^{\#k}(S) = \sum_j \bar{\chi}_j(k-S)^j$ then gives $0 = \int \chi \chi^{\#k} dm_t = \int |\chi|^2 dm_t$, contradicting the positive original Gram of χ . Thus $\mathcal{B}(t) > 0$. RMT11 follows from its complete original angle proof and concavity of $\sqrt{1-e^{-z}}$ on the retained a_q -entry scalar list. Applying all five inequalities to this same derivative proves $|\Delta\mathcal{B}| \leq \mathcal{J}_{h,k}^{(5)} \leq \mathcal{J}_{h,k}$. Compactness and the positive continuous $\mathcal{B}$ make every denominator in (AT18b) nonzero and bounded away from zero.

For each retained relative eigenvalue,

$$\int_0^1 c_j(t) dt = \log b_j, \quad \int_0^1 d(t) dt = \log(b_n/b_1) = \log \kappa_*. \quad (\text{AT21})$$

This follows by differentiating $\log(1 - t + tb_j)$; it includes $b_j = 1$, whose integrand is zero. Thus $\int S_{\text{AW}} = \mathcal{L}_q(D_*)$ and $\mathcal{J}_{h,k} \leq \int S_{\text{SP}}$. Each ratio in $\mathcal{L}_q$ is at most κ_* ; the retained total coefficient is $1 + 2(q-1) = 2q-1$. This proves all inequalities. Finally the absolute branches of (AT19) follow by applying both signs of (AT18), and its constructed signed branches follow from (AT19bb) proved below, to the exact equality $\mathcal{B}_{h,k}^{\text{ar}} = \mathcal{B}_{h,k}^{\Gamma} + \Delta\mathcal{B}$. $\square$

For every integer $\ell_N \geq 0$ the constructed signed approximation is applied on these exact endpoint forms as well. Here $M_z = M_*(t)|_{t=z}$, $C_z = C(t)|_{t=z}$, $A_z = (U(t) - W(t))|_{t=z}$, and $D = D_*$. Thus the RMT/ISM coefficient map is the identity with $x_{\text{RMT}} = z = t$, and its $\mathcal{B}$ is (AT14). Write $D = (M_0)^{-1}M_1$, with its complete ordered positive eigenvalues $b_1, \dots, b_{2q+1}$, and let A_z be this same signed projection difference. Define

$$\begin{aligned} \alpha_z^\circ &= \frac{2(1-z) + z(b_1 + b_{2q+1})}{2}, \quad H_z^\circ = I - \frac{(1-z)I + zD}{\alpha_z^\circ}, \\ C_z^{[\ell_N]} &= (\alpha_z^\circ)^{-1} \sum_{r=0}^{\ell_N} (H_z^\circ)^r (D - I), \quad R_z^{[\ell_N]} = (H_z^\circ)^{\ell_N+1} C_z, \\ E_z^{[\ell_N]} &= R_z^{[\ell_N]} - \frac{\text{Tr } R_z^{[\ell_N]}}{2q+1} I. \end{aligned} \quad (\text{AT19ba})$$

The symbols $\alpha_z^\circ, H_z^\circ$ preserve every original source parameter and relation map; ℓ_N is only the finite series truncation. Multiplication of the finite geometric sum gives $C_z - C_z^{[\ell_N]} = R_z^{[\ell_N]}$. All factors are real functions of the positive relative operator D , hence self-adjoint in the original $M_z = M_0((1-z)I + zD)$ metric. Since $\text{Tr } A_z = 0$, the derivative of the same four-volume function is its signed center $\text{Tr}(C_z^{[\ell_N]} A_z)$ plus $\text{Tr}(E_z^{[\ell_N]} A_z)$. Consequently

$$\begin{aligned} j_{\ell_N} &= \int_0^1 \text{Tr}(C_z^{[\ell_N]} A_z) dz, \\ e_{\ell_N} &= \int_0^1 \sqrt{\text{Tr}((E_z^{[\ell_N]})^2) \text{Tr}(A_z^2)} dz, \\ j_{\ell_N} - e_{\ell_N} &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_{\ell_N} + e_{\ell_N}. \end{aligned} \quad (\text{AT19bb})$$

For this block $\mathcal{B}$ denotes precisely the four-volume function identified above. The signed inequalities follow by Hilbert–Schmidt Cauchy–Schwarz in that unchanged metric and integration of the exact derivative. The full residual calculation ISM29–37, equivalently RMT19–21, uses $\vartheta = (b_{2q+1} - b_1)/(b_{2q+1} + b_1) < 1$ and $K_D = \sum_j (|b_j - 1|/\min\{1, b_j\})^2$ to prove

$$0 \leq e_{\ell_N} \leq \vartheta^{\ell_N+1} \sqrt{K_D(8q-6)} \longrightarrow 0. \quad (\text{AT19bc})$$

Indeed every eigenvalue of H_z° has modulus at most ϑ , centering the residual subtracts a nonnegative square from its squared Hilbert–Schmidt norm, and $\text{Tr}(A_z^2) \leq 8q-6$ by the full flag spectra and the common range. This proves the bound with all eigenvalue multiplicities retained. The displayed integrals are exact; a numerical evaluation must retain its own integration error. Intersecting this proved signed interval with the already proved absolute and nonlinear intervals supplies the additional endpoint entries below, for every specified ℓ_N .

The same actual-moment calculation also propagates to the nonlinear endpoint coordinate. Set $\mathcal{H}_{a_q}(B) = 2 \text{arcosh}(e^{B/(2a_q)})$ and $H_0 = \mathcal{H}_{a_q}(\mathcal{B}_{h,k}^\Gamma)$. Its exact derivative is $\mathcal{H}_{a_q}'(B) = 1/(a_q \sqrt{1 - e^{-B/a_q}})$ and its continuous inverse is $2a_q \log \cosh(z/2)$ for $z \geq 0$. The

positivity just proved permits the chain rule throughout this fixed path. Dividing the five derivative bounds by their positive factor and integrating gives $|\mathcal{H}_{a_q}(\mathcal{B}_{h,k}^{\text{ar}}) - H_0| \leq \mathcal{I}_{h,k} \leq \log \kappa_*$. Consequently the existing interval (AT19) has the simultaneous strengthening

$$\begin{aligned} \mathcal{B}_{h,k}^{\text{ar}} &\geq \max \left\{ 0, \mathcal{B}_{h,k}^\Gamma - \mathcal{J}_{h,k}^{(5)}, 2a_q \log \cosh \frac{\max\{0, H_0 - \mathcal{I}_{h,k}\}}{2}, \mathcal{B}_{h,k}^\Gamma + j_{\ell_N} - e_{\ell_N} \right\}, \\ \mathcal{B}_{h,k}^{\text{ar}} &\leq \min \left\{ \mathcal{B}_{h,k}^\Gamma + \mathcal{J}_{h,k}^{(5)}, 2a_q \log \cosh \frac{H_0 + \mathcal{I}_{h,k}}{2}, \mathcal{B}_{h,k}^\Gamma + j_{\ell_N} + e_{\ell_N} \right\}. \end{aligned} \quad (\text{AT19a})$$

Every inverse and sign is the original RMT16–RMT18 calculation on the forms already specified in AT; the scalar continuous value at zero bounds an actual strictly positive endpoint.

When $M_*(1) = cM_*(0)$, every $b_j = c$, so $d = 0$ and $\mathcal{J}_{h,k} = 0$. The original quotient determinants are $(1 - t + tc)^q V_N(0)$, whose four factors cancel exactly in (AT14). This retains both original masses. For an invertible coefficient arrow $F : H' \rightarrow H$, let $M'_* = F^* M_* F$ and transport every subspace by F^{-1} . The projector formula gives $P'_N = F^{-1} P_N F$, $Q'_N = F^{-1} Q_N F$, $D'_* = F^{-1} D_* F$, and $C' = F^{-1} C F$. Their spectra, intersection dimensions, $\text{Tr}(UW)$, the signed trace and $\mathcal{J}_{h,k}$ consequently agree under this exact map.

For the original reflection-stable quartet with displacement $\delta > 0$ and multiplicity m , retain $q = [1 + k(m - 1)](k + 1)^2$. Its supplied Gamma theorem gives $\mathcal{B}_{h,k}^\Gamma \geq 4q \log(\delta k / (2\sqrt{5}))$. Applying (AT19) with its actual sign proves

$$\begin{aligned} \mathcal{B}_{h,k}^{\text{ar}} &\geq 4q \log \left(\frac{\delta k}{2\sqrt{5}} \right) - \mathcal{J}_{h,k}^{(5)} \\ &\geq 4q \log \left(\frac{\delta k}{2\sqrt{5}} \right) - \mathcal{L}_q(D_*) \\ &\geq 4q \log \left(\frac{\delta k}{2\sqrt{5}} \right) - (2q - 1) \log \kappa_*, \\ \log \kappa_* &\geq \frac{4q \log(\delta k / (2\sqrt{5})) - \mathcal{B}_{h,k}^{\text{ar}}}{2q - 1}. \end{aligned} \quad (\text{AT22})$$

The signed branch also yields $\mathcal{B}_{h,k}^{\text{ar}} \geq 4q \log(\delta k / (2\sqrt{5})) + j_{\ell_N} - e_{\ell_N}$: substitute the same Gamma lower bound into $\mathcal{B}_{h,k}^{\text{ar}} \geq \mathcal{B}_{h,k}^\Gamma + j_{\ell_N} - e_{\ell_N}$. This branch and (AT22) are both retained by their maximum. The last denominator is positive for every $q \geq 1$. This is the same actual packet application, with its earlier necessary condition updated to the stronger proved finite control. No existence of such a quartet or uniform growth bound is assumed. The full signed formulas (AT15)–(AT17), all source primitives and the retained cohomology observations remain the objects whose moments and actual overlaps enter the bound.

1.5 Source dependencies and precise status

The complete retained Tau Base note supplies (AT1), its tensor pushforward, right adjoint and supported maps. The original arithmetic-input and cyclic-sum proofs supply (AT2), full packet units and the theta primitives. The Gamma/source-density notes supply the actual measures and finite moments in (AT4)–(AT5). The new Deligne-bound note supplies the full quartet Gamma lower bound used only in (AT22). Each is preserved in the accompanying source closure, with its actual edition and hash.

The new proved calculations here are (AT6)–(AT21) on those unchanged input objects, and their substitution (AT22). They give an analytic calculation on the canonical finite complex mapping to $A_{\tau,k}(\mathcal{T}^{\boxtimes k})$. They neither replace that pushforward by a generic stalk nor deduce a verdict about the entire tau-based program from an auxiliary model.

2 Exact relation-window spectral transport on the original tau complex

This calculation continues arithmetic determinant transport on its original finite source over the absolute tau base. It computes the full spectra of the two retained window operators, constructs an isometry intertwining them in the actual source metric, and strengthens the signed determinant bound. The source measures, their masses, the cyclic polynomial, the Taylor unit, and the cohomology observation remain fixed.

2.1 The original source and its four windows

Use a full actual packet h for $g = 2\xi$, an integer $k \geq 1$, and the original cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$. Let $E = \mathbb{C}[S]/(\chi)$, $E_h = \mathbb{C}[s]/(h)$, and $v_h = j_h(g/h) \in E_h^\times$. The original tensor-sum observation is

$$\eta : E \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_h^{(k)})1. \quad (\text{AW1})$$

The original section $\sigma_h : E_h \hookrightarrow Q = \mathcal{B}/\Theta V$ gives the separate injection $\sigma_h^{\otimes k} \eta : E \hookrightarrow Q^{\otimes k}$. Here $Q^{\otimes k}$ is the top cohomology of the original tensor pushforward over $F_{1,\tau} = \{\tau, 1\}$; the one-factor complex is $[V \oplus V \xrightarrow{\Theta(\phi-\hat{\psi})} \mathcal{B}]$. The complete Tau Base and cyclic source proofs accompany the preceding arithmetic determinant transport calculation (AT).

For $N \geq q - 1$, retain the increasing-power spaces and maps

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N = \times \chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0, \quad \mathcal{P}_{-1} = 0. \quad (\text{AW2})$$

The common source is $H = \mathcal{P}_{2q}$, of dimension $n = 2q + 1$. Its original coordinate on the integration line is $S = k/2 + iu$. Set $v_N(u) = (1, S(u), \dots, S(u)^N)$ and retain the two densities

$$\begin{aligned} m_1(u) &= w_h^{*k}(u), & w_h(u) &= \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \\ m_0(u) &= \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, & c_a &= 2^{1-2a} \Gamma(2a). \end{aligned} \quad (\text{AW3})$$

Their actual masses are μ_h^k and $(2\pi)^{k/2}$. For the real parameter $0 \leq t \leq 1$, put $m_t = (1-t)m_0 + tm_1$, $M_N(t) = \int v_N^* v_N m_t$, and $M(t) = M_{2q}(t)$. The source density proofs give all finite moments and positivity almost everywhere, so all these finite Grams are positive definite. We use the original form $\langle x, y \rangle_t = x^* M(t) y$ on H .

Let $P_N(t)$ be the orthogonal projection onto $\mathcal{P}_N \subset H$, and $Q_N(t)$ the projection onto $\chi \mathcal{P}_{N-q} \subset H$. Set $Q_{q-1} = 0$. For the coefficient inclusion $I_N : \mathcal{P}_N \rightarrow H$, their fully typed coefficient formulas are

$$\begin{aligned} P_N &= I_N M_N^{-1} I_N^* M, \\ Q_N &= I_N B_N (B_N^* M_N B_N)^{-1} B_N^* I_N^* M. \end{aligned} \quad (\text{AW4})$$

The zero relation domain at $N = q - 1$ contributes the zero operator. Let $G_N(t)$ be the minimum-lift Gram on E from (AW2) and $V_N(t) = \det G_N(t)$. Retain

$$\begin{aligned} \mathcal{B}(t) &= \log \frac{V_{q-1}(t) V_q(t)}{V_{2q-1}(t) V_{2q}(t)}, \\ C(t) &= M(t)^{-1} (M(1) - M(0)), \\ U(t) &= (Q_{2q-1} - Q_{q-1}) + (Q_{2q} - Q_q), \\ W(t) &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q). \end{aligned} \quad (\text{AW5})$$

The AT proof, included in full, establishes $V_N = \det M_N / \det(B_N^* M_N B_N)$ with the original orientation $(-1)^{q(N-q+1)}$, and

$$\Delta \mathcal{B} = \mathcal{B}(1) - \mathcal{B}(0) = \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt. \quad (\text{AW6})$$

Thus the following computation evaluates the operators in the already proved signed arithmetic correction.

2.2 Full spectra and an explicit source isometry

Theorem 2.1. *In the original $M(t)$ metric, both $U(t)$ and $W(t)$ have spectrum*

$$0 \text{ with multiplicity } q, \quad 1 \text{ with multiplicity } 2, \quad 2 \text{ with multiplicity } q - 1. \quad (\text{AW7})$$

The last multiplicity is zero when $q = 1$. There is an explicitly constructed smooth $M(t)$ -unitary $T(t) : H \rightarrow H$ such that $U(t) = T(t)W(t)T(t)^{-1}$.

Proof. Every subspace in either of the following flags is the literal displayed polynomial subspace, with its original coefficient inclusion. For the relation flag use

$$F_0 = \chi \mathcal{P}_0 \subset F_1 = \chi \mathcal{P}_{q-1} \subset F_2 = \chi \mathcal{P}_q \subset H. \quad (\text{AW8})$$

Their dimensions are $1, q, q+1$, since multiplication by χ is injective. When $q = 1$, $F_0 = F_1$ is retained as that equality. Projections of nested subspaces commute and their products equal the projection of the smaller subspace: each larger space is the orthogonal direct sum of the smaller space and its orthogonal complement within the larger space. On the original orthogonal decomposition $F_0 \oplus (F_1 \cap F_0^\perp) \oplus (F_2 \cap F_1^\perp) \oplus F_2^\perp$, the operator $U = Q_{F_1} + Q_{F_2} - Q_{F_0}$ acts respectively by $1, 2, 1, 0$. The dimensions are respectively $1, q - 1, 1, q$.

For W use $\mathcal{P}_{q-1} \subset \mathcal{P}_q \subset \mathcal{P}_{2q-1} \subset \mathcal{P}_{2q} = H$. On the four corresponding orthogonal pieces, W acts by $0, 1, 2, 1$, with dimensions $q, 1, q - 1, 1$. This proves (AW7), including the repeated middle subspaces at $q = 1$.

For an explicit intertwiner retain every Gram factor as follows. For any specified ordered independent list $z_0, \dots, z_d$, apply the recursion

$$y_j = z_j - \sum_{i < j} e_i \langle e_i, z_j \rangle_t, \quad a_j = (y_j^* M(t) y_j)^{1/2} > 0, \quad e_j = y_j / a_j. \quad (\text{AW9})$$

The positive real square root fixes its phase. No source density or mass is changed; each a_j is the norm in the original form. Independence proves $y_j \neq 0$ at every step, so this recursion is defined and smooth on the entire parameter interval.

First use $z_j = S^j$, $0 \leq j \leq 2q$, obtaining $p_0, \dots, p_{2q}$. Next use $z_j = \chi S^j$, $0 \leq j \leq q$, obtaining $r_0, \dots, r_q$. These lie in the successive relation spaces in (AW8). Finally let $R_{2q}(t) : E \rightarrow H$ be the original minimum section from AT6 and apply (AW9) to $R_{2q}(t)[1], \dots, R_{2q}(t)[S^{q-1}]$, obtaining $c_0, \dots, c_{q-1}$. The relation orthogonality in AT6 puts this entire list in $F_2^\perp$. Since $J_{2q} R_{2q} = I_E$, the list is independent and spans that q -dimensional complement.

Define T on the first orthonormal basis by

$$\begin{aligned} Tp_j &= c_j & (0 \leq j \leq q-1), \\ Tp_q &= r_0, & Tp_{2q} = r_q, \\ Tp_{q+j} &= r_j & (1 \leq j \leq q-1). \end{aligned} \quad (\text{AW10})$$

The middle range is empty for $q = 1$, with p_q and p_{2q} still distinct. The target list is an orthonormal basis. Thus this linear map is invertible and $T^*MT = M$. The spectral actions already calculated agree on each indicated basis vector, proving $UT = TW$. The recursion, the smooth minimum section and the positive norms prove the asserted smoothness. This completes the construction. $\square$

The exact isometry gives a commutator form of the signed integrand:

$$\text{Tr}((U - W)C) = \text{Tr}(WT^{-1}[C, T]), \quad [C, T] = CT - TC. \quad (\text{AW11})$$

Indeed cyclicity moves T in $\text{Tr}(TWT^{-1}C)$ to the right, and subtraction then gives the displayed product. This T is an isometry of the original source; its cohomology observation is computed, rather than silently assigned an intertwining property. The unchanged arithmetic observation $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} : H \rightarrow Q^{\otimes k}$ satisfies

$$\mathcal{O}T = \sigma_h^{\otimes k} \eta J_{2q} T, \quad \mathcal{O}T - \mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} (T - I), \quad \ker(\mathcal{O}T) = T^{-1}F_2. \quad (\text{AW12})$$

The last identity follows from both injections in (AW1) and $\ker J_{2q} = F_2$. By (AW10), $T^{-1}F_2$ is exactly $\text{span}\{p_q, \dots, p_{2q}\} = \mathcal{P}_{q-1}^\perp$ in H . At any fixed nonempty support label A , lift every displayed linear map by $(A, x) \mapsto (A, fx)$ and send τ to τ . Consequently the supported-zero inverse image of the lifted $\mathcal{O}T$ at A is the full set $\{A\} \times \mathcal{P}_{q-1}^\perp$. The Taylor unit in (AW1) and the defect in (AW12) remain present.

2.3 The spectral bound with every eigenvalue retained

Let $b_1 \leq \dots \leq b_n$ be all eigenvalues, counted with multiplicity, of $D = M(0)^{-1}M(1)$ on its original $M(0)$ Hilbert space. All are positive. Define

$$\mathcal{L}_q(D) = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}, \quad \kappa(D) = b_n/b_1. \quad (\text{AW13})$$

For $q = 1$ the sum is empty and $\mathcal{L}_1(D) = \log(b_3/b_1)$.

Retain also the actual polynomial-space overlap. At each t , let $s(t) = \dim(\text{ran } U \cap \text{ran } W)$ and $d(t) = c_n(t) - c_1(t)$. With the actual $c_j(t)$ of (AW17), set

$$\begin{aligned} S_{\text{AW}}(t) &= c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j), \\ S_{\text{SP}}(t) &= \frac{d(t)}{2} \min \left\{ 4q - 2s(t), \sqrt{(2q+1)(8q-4-2\text{Tr}(UW))} \right\}, \\ \mathcal{J}_{h,k} &= \int_0^1 \min\{S_{\text{AW}}(t), S_{\text{SP}}(t)\} dt. \end{aligned} \quad (\text{AW13a})$$

The overlap refinement from SP8–SP10 is transported to this earlier calculation by the exact arrows $x = t, y = u, B_N^{\text{SP}} = I_N B_N^{\text{AW}}$. Indeed $M_N = I_N^* M I_N$ gives $(B_N^{\text{SP}})^* M B_N^{\text{SP}} = B_N^* M_N B_N$, so its projectors are precisely (AW4), and $(\mathcal{R}^{\text{SP}}, \mathcal{P}^{\text{SP}}, T^{\text{SP}}) = (U, W, C)$. The original Gamma densities agree because $r_{1/4}^{*k} = c_{1/4}^k r_{k/4}/c_{k/4}$ with $c_{1/4} = \sqrt{2\pi}$, exactly (AW3). The arithmetic densities already coincide as w_h^{*k} . Thus this comparison preserves the moments, masses, polynomial coefficients and all four endpoint signs.

The exact trace norm and the centered Hilbert–Schmidt estimate are also retained before taking a minimum. On this same source define

$$\begin{aligned} S_{\text{tr}}(t) &= \frac{d(t)}{2} \|U(t) - W(t)\|_1, \\ \Sigma(t) &= \text{Tr}(C(t)^2) - \frac{(\text{Tr } C(t))^2}{2q+1}, \\ S_{\text{HS}}(t) &= \sqrt{\Sigma(t) \text{Tr}((U(t) - W(t))^2)}. \end{aligned} \quad (\text{AW13c})$$

The trace-norm inequality follows by subtracting the midpoint of the extreme eigenvalues from the metric derivative, since the trace of the signed projection difference is zero. In an orthonormal eigenbasis of that difference, every diagonal derivative entry is bounded in modulus by half the spectral diameter; summing with the absolute eigenvalues proves the assertion. For the other bound subtract the trace mean $\text{Tr}(C(t))I/(2q+1)$ instead. The square of its Hilbert–Schmidt norm in the unchanged

source metric is exactly $\Sigma(t)$. Hilbert–Schmidt Cauchy–Schwarz with the signed difference proves that its trace pairing is bounded by $S_{\text{HS}}(t)$. Positivity of both factors under the square root follows from their self-adjointness. Thus both functions bound the same absolute signed derivative on general positive source forms; the angle term below still uses the specified moment measures. These are RMT10a–10b and ISM25–26 through the already proved original coefficient and projection maps.

The original-moment angle calculation supplies a further simultaneous bound at this same claim site. Put $a_q = 2q - 1$ and

$$\begin{aligned} A_{h,k}(t) &= a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}} d(t), \\ \mathcal{J}_{h,k}^{(5)} &= \int_0^1 \min\{S_{\text{AW}}(t), S_{\text{tr}}(t), S_{\text{SP}}(t), S_{\text{HS}}(t), A_{h,k}(t)\} dt, \\ \mathcal{I}_{h,k} &= \int_0^1 \min \left\{ d(t), \frac{S_{\text{AW}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}}, \frac{S_{\text{SP}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}}, \right. \\ &\quad \left. \frac{S_{\text{tr}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}}, \frac{S_{\text{HS}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}} \right\} dt. \end{aligned} \tag{AW13b}$$

These specialize RMT1–RMT18 through $a_{\text{RMT}} = a_q$, $x = t$, $C_x = C(t)$ and $(U_x, W_x) = (U(t), W(t))$. The density and relation arrows were proved in (AW3)–(AW4) and (AW13a), so every source moment and original cohomology map agrees. This application uses the actual moment measures (AW3). It does not impose moment structure on an arbitrary positive coefficient Gram.

Theorem 2.2. *On the unchanged arithmetic source,*

$$|\Delta \mathcal{B}| \leq \mathcal{J}_{h,k}^{(5)} \leq \mathcal{J}_{h,k} \leq \mathcal{L}_q(D) \leq (2q - 1) \log \kappa(D). \tag{AW14}$$

The previously proved coefficient $2q - 1$ and full relative-spectrum bound are retained. The pointwise minimum additionally preserves the actual polynomial-space overlap, and is at most $\int_0^1 S_{\text{SP}}(t) dt$.

Proof. We first prove the needed finite spectral trace inequality. Let C be self-adjoint in a finite Hilbert space with eigenvalues $c_1 \leq \dots \leq c_n$. If P is an orthogonal projection of rank r , let e_j be an orthonormal eigenbasis of C and $d_j = \langle e_j, P e_j \rangle$. Then $0 \leq d_j \leq 1$ and $\sum_j d_j = r$, since $d_j = \|P e_j\|^2$ and $\text{Tr } P = r$. Thus $\text{Tr}(PC) = \sum_j c_j d_j$. Its upper bound by $\sum_{j=n-r+1}^n c_j$ follows by subtracting the two expressions: the mass $\sum_{j \leq n-r} d_j$ equals $\sum_{j > n-r} (1 - d_j)$, and every coefficient in the latter group is at least every coefficient in the former. Pairing the same nonnegative mass, or bounding both groups by c_{n-r+1} , proves the inequality. Applying this argument to $-C$ proves the lower bound $\sum_{j=1}^r c_j$. This proof includes repeated eigenvalues and the ranks zero and n .

For an operator Z with spectrum (AW7), its spectral projection A_Z onto eigenvalues 1, 2 has rank $q + 1$, and its spectral projection B_Z onto eigenvalue 2 has rank $q - 1$. The exact decomposition is $Z = A_Z + B_Z$. Applying the proved projection bounds gives

$$\sum_{j=1}^{q+1} c_j + \sum_{j=1}^{q-1} c_j \leq \text{Tr}(ZC) \leq \sum_{j=q+1}^{2q+1} c_j + \sum_{j=q+3}^{2q+1} c_j. \tag{AW15}$$

Each bound is attained for an abstract operator of this spectrum by assigning its eigenspaces to the appropriate eigenvectors of C . This attainment statement is only about the displayed spectral constraint; it assigns no freely chosen orientation to the actual polynomial relation spaces. Both U and W satisfy (AW15). Subtract their bounds and cancel the common c_{q+1} term only after writing both complete sums:

$$|\text{Tr}((U - W)C)| \leq c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j). \tag{AW16}$$

No mutual commutation of U, W, C is used.

The range dimensions in (AW7) give $s(t) \geq 1$. Let E_L be the orthogonal projector onto their intersection. Each $Z = U, W$ is at least its range projection, since its nonzero eigenvalues are 1, 2, and therefore $Z - E_L$ is positive of trace $2q - s(t)$. Writing $A = U - W = (U - E_L) - (W - E_L)$ proves

$$\|A\|_1 \leq 4q - 2s(t), \quad \text{Tr } A = 0, \quad \text{Tr } A^2 = 8q - 4 - 2 \text{Tr}(UW), \quad \|A\|_1 \leq \sqrt{(2q+1) \text{Tr } A^2}. \quad (\text{AW16a})$$

The square identity uses $\text{Tr } U^2 = \text{Tr } W^2 = 4q - 2$ and cyclicity. The last inequality is Cauchy-Schwarz on every real eigenvalue of the self-adjoint A . Subtracting $(c_1 + c_n)I/2$ from C does not change its trace pairing with A . In an orthonormal eigenbasis of A , that pairing is bounded in absolute value by $d(t)\|A\|_1/2$. This proves $|\text{Tr}((U - W)C)| \leq S_{\text{SP}}(t)$ in addition to (AW16), and hence bounds it by their pointwise minimum. Smooth positive Grams give bounded continuous eigenvalues and overlaps; $s(t)$ is Borel because its rank strata are specified by matrix minors. Therefore the minimum is integrable. The signed identity (AW6) now proves $|\Delta\mathcal{B}| \leq \mathcal{J}_{h,k}$ and $\mathcal{J}_{h,k} \leq \int S_{\text{SP}}$. For the original moment path, the exact original-source identification above also applies RMT11-RMT12: its full crossed angle lists give $|\mathcal{B}'| \leq A_{h,k}$, and $\mathcal{B} > 0$ follows because otherwise $1 \perp \chi\mathcal{P}_q$ would force $\int |\chi|^2 m_t = 0$ by the reflected polynomial $\chi^{\#k}(S) = \sum_j \overline{\chi_j}(k - S)^j$. Taking the pointwise minimum of this additional proved inequality with (AW16), (AW16a), and both complete bounds in (AW13c) gives $|\Delta\mathcal{B}| \leq \mathcal{J}_{h,k}^{(5)}$. The complete proof of the original angle estimate and its exact chain-rule continuation is retained in EP and RMT; it uses AW's earlier projection identities, not the final claim (AW14).

For the actual interpolated metric, $C(t) = ((1 - t)I + tD)^{-1}(D - I)$, so its eigenvalues in increasing order are

$$c_j(t) = \frac{b_j - 1}{1 - t + tb_j}. \quad (\text{AW17})$$

The denominator is positive and the derivative with respect to b_j is $(1 - t + tb_j)^{-2} > 0$. Thus the ordering is valid for the entire closed interval. Each $c_j(t)$ is the derivative of $\log(1 - t + tb_j)$ and has integral $\log b_j$. Integrating (AW16) proves $\mathcal{J}_{h,k} \leq \mathcal{L}_q(D)$; the preceding pointwise-minimum argument proves the first inequality of (AW14). Every ratio in (AW13) lies between 1 and b_n/b_1 , so retaining its coefficient gives $1 + 2(q - 1) = 2q - 1$ times $\log \kappa(D)$ for the second bound. This proves the theorem for all $q \geq 1$. $\square$

All bounds retain the literal source masses. If $M(1) = cM(0)$, every $b_j = c$, so (AW13) is zero. Directly each original quotient determinant is multiplied at time t by $(1 - t + tc)^q$, and the four displayed factors in (AW5) cancel. For a fixed invertible source-coordinate arrow F , the transported forms are $M' = F^*MF$, all subspaces have their F^{-1} images, and $D' = F^{-1}DF$. Hence the eigenvalues, (AW6) and (AW13) are unchanged under this explicit map.

For every integer $\ell_N \geq 0$ the constructed signed approximation is applied on these exact end-point forms as well. Here $M_z = M(t)|_{t=z}$, $C_z = C(t)|_{t=z}$ and $A_z = (U(t) - W(t))|_{t=z}$, with $D = D$. The exact RMT/ISM source arrow is the identity, and $\mathcal{B}$ is the function (AW5). Write $D = (M_0)^{-1}M_1$, with its complete ordered positive eigenvalues $b_1, \dots, b_{2q+1}$, and let A_z be this same signed projection difference. Define

$$\begin{aligned} \alpha_z^\circ &= \frac{2(1 - z) + z(b_1 + b_{2q+1})}{2}, & H_z^\circ &= I - \frac{(1 - z)I + zD}{\alpha_z^\circ}, \\ C_z^{[\ell_N]} &= (\alpha_z^\circ)^{-1} \sum_{r=0}^{\ell_N} (H_z^\circ)^r (D - I), & R_z^{[\ell_N]} &= (H_z^\circ)^{\ell_N+1} C_z, \\ E_z^{[\ell_N]} &= R_z^{[\ell_N]} - \frac{\text{Tr } R_z^{[\ell_N]}}{2q+1} I. \end{aligned} \quad (\text{AW17aa})$$

The symbols $\alpha_z^\circ, H_z^\circ$ preserve every original source parameter and relation map; ℓ_N is only the finite series truncation. Multiplication of the finite geometric sum gives $C_z - C_z^{[\ell_N]} = R_z^{[\ell_N]}$. All factors are

real functions of the positive relative operator D , hence self-adjoint in the original $M_z = M_0((1 - z)I + zD)$ metric. Since $\text{Tr } A_z = 0$, the derivative of the same four-volume function is its signed center $\text{Tr}(C_z^{[\ell_N]} A_z)$ plus $\text{Tr}(E_z^{[\ell_N]} A_z)$. Consequently

$$\begin{aligned} j_{\ell_N} &= \int_0^1 \text{Tr}(C_z^{[\ell_N]} A_z) dz, \\ e_{\ell_N} &= \int_0^1 \sqrt{\text{Tr}((E_z^{[\ell_N]})^2) \text{Tr}(A_z^2)} dz, \\ j_{\ell_N} - e_{\ell_N} &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_{\ell_N} + e_{\ell_N}. \end{aligned} \tag{AW17ab}$$

For this block $\mathcal{B}$ denotes precisely the four-volume function identified above. The signed inequalities follow by Hilbert–Schmidt Cauchy–Schwarz in that unchanged metric and integration of the exact derivative. The full residual calculation ISM29–37, equivalently RMT19–21, uses $\vartheta = (b_{2q+1} - b_1)/(b_{2q+1} + b_1) < 1$ and $K_D = \sum_j (|b_j - 1|/\min\{1, b_j\})^2$ to prove

$$0 \leq e_{\ell_N} \leq \vartheta^{\ell_N+1} \sqrt{K_D(8q - 6)} \longrightarrow 0. \tag{AW17ac}$$

Indeed every eigenvalue of H_z° has modulus at most ϑ , centering the residual subtracts a nonnegative square from its squared Hilbert–Schmidt norm, and $\text{Tr}(A_z^2) \leq 8q - 6$ by the full flag spectra and the common range. This proves the bound with all eigenvalue multiplicities retained. The displayed integrals are exact; a numerical evaluation must retain its own integration error. Intersecting this proved signed interval with the already proved absolute and nonlinear intervals supplies the additional endpoint entries below, for every specified ℓ_N .

The original quartet Gamma theorem, when applied to the actual full quartet packet with displacement $\delta > 0$ and common order m , retains $q = [1 + k(m - 1)](k + 1)^2$ and gives

$$\mathcal{B}_{h,k}^{\text{ar}} \geq 4q \log\left(\frac{\delta k}{2\sqrt{5}}\right) + \max\{-\mathcal{J}_{h,k}^{(5)}, j_{\ell_N} - e_{\ell_N}\}. \tag{AW18}$$

This substitutes the complete supplied Gamma theorem into the lower branches of (AW14) and the constructed signed inequality (AW17ab), then takes their maximum. It retains the stronger actual overlap bound, and also implies the earlier lower bounds with $-\mathcal{L}_q(D)$ and $-(2q - 1) \log \kappa(D)$ respectively. It proves no uniform estimate for the eigenvalues b_j , and asserts no existence of an off-line packet. The new computed objects are the exact window spectra, their source isometry, its arithmetic observation and commutator, and the strengthened finite bound.

3 The one-dimensional quotient: exact angle and nonlinear transport

When $q = 1$, the same four endpoints are 0, 1, 1, 2. Their middle factors cancel in the specified ratio, leaving $\mathcal{B}(t) = \log(V_0(t)/V_2(t))$ with both original masses still present. Write $\chi = S - a$. The actual operators now are $U = Q_2$ and $W = I - P_0$. If $\mathcal{N} = (\chi \mathcal{P}_1)^\perp$ and $P_{\mathcal{N}} = I - Q_2$, then $U - W = P_0 - P_{\mathcal{N}}$. Both projections have rank one.

Let $e = 1 \in H$, $z = R_2(t)[1]$, and retain their actual norms $\langle e, e \rangle_t = V_0(t)$ and $\langle z, z \rangle_t = V_2(t)$. The source splitting gives $z = P_{\mathcal{N}}e$, since $J_2e = 1$. Thus $\langle e, z \rangle_t = V_2(t)$, and the angle between the two lines has the exact value

$$\cos^2 \theta(t) = \frac{|\langle e, z \rangle_t|^2}{\langle e, e \rangle_t \langle z, z \rangle_t} = \frac{V_2(t)}{V_0(t)} = e^{-\mathcal{B}(t)}. \tag{AW19}$$

Both Grams are positive, so the cosine is positive. It is strictly less than one for the actual measure (AW3). Otherwise $e \in \mathcal{N}$, implying $\int \chi(S) m_t = 0$ and $\int S \chi(S) m_t = 0$. On the unchanged line $\bar{S} = k - S$, and consequently $|\chi(S)|^2 = ((k - \bar{a}) - S) \chi(S)$. The two equalities would force $\int |\chi(S)|^2 m_t = 0$,

contrary to the positive Gram of this nonzero polynomial. Hence $0 < \theta(t) < \pi/2$ and $\mathcal{B}(t) > 0$ on the whole closed interval.

For completeness, take the unit vector $p = e/\sqrt{V_0}$ and write $z/\sqrt{V_2} = \cos \theta p + \sin \theta r$, where r is the resulting unit vector orthogonal to p . The phase is already fixed since $\langle e, z \rangle_t = V_2 > 0$. In this orthonormal pair the exact difference of projections is

$$(P_0 - P_N)|_{\text{span}(p,r)} = \begin{pmatrix} \sin^2 \theta & -\sin \theta \cos \theta \\ -\sin \theta \cos \theta & -\sin^2 \theta \end{pmatrix}. \quad (\text{AW20})$$

Its trace is zero and its determinant is $-\sin^2 \theta$. It vanishes on the remaining one-dimensional orthogonal complement. Thus its full eigenvalues are $\sin \theta, 0, -\sin \theta$. Applying the proved rank-one spectral trace bounds to its positive and negative eigenprojections yields

$$|\dot{\mathcal{B}}(t)| \leq \sqrt{1 - e^{-\mathcal{B}(t)}} (c_3(t) - c_1(t)). \quad (\text{AW21})$$

This retains the actual alignment of the two lines.

The function $\Phi(B) = 2 \operatorname{arcosh}(e^{B/2})$ for $B > 0$ has derivative $\Phi'(B) = 1/\sqrt{1 - e^{-B}}$, by differentiating the inverse of the hyperbolic cosine on its positive branch. The preceding strict positivity on the compact interval justifies its smooth composition with $\mathcal{B}(t)$. Divide (AW21) by its positive square-root factor. The four independent generic inequalities (AW16), (AW16a), and the trace-norm and Hilbert–Schmidt inequalities (AW13c) can all be divided by that same factor because $q = 1$ gives $a_q = 1$ and $\mathcal{B} > 0$. Their pointwise minimum integrates to $\mathcal{I}_{h,k}$ in (AW13b); integration of (AW17) bounds it by $\log(b_3/b_1)$. The exact nonlinear endpoint bound is

$$\left| 2 \operatorname{arcosh}(e^{\mathcal{B}(1)/2}) - 2 \operatorname{arcosh}(e^{\mathcal{B}(0)/2}) \right| \leq \mathcal{I}_{h,k} \leq \log(b_3/b_1). \quad (\text{AW22})$$

Combining (AW22) with the signed interval (AW17ab), with the original values $A_0 = \operatorname{arcosh}(e^{\mathcal{B}(0)/2})$ and $L = \mathcal{I}_{h,k}/2 \leq \frac{1}{2} \log(b_3/b_1)$, monotonicity of $2 \log \cosh A$ for $A \geq 0$ gives

$$\begin{aligned} \mathcal{B}(1) &\geq \max\{2 \log \cosh(\max\{0, A_0 - L\}), \mathcal{B}(0) + j_{\ell_N} - e_{\ell_N}\}, \\ \mathcal{B}(1) &\leq \min\{2 \log \cosh(A_0 + L), \mathcal{B}(0) + j_{\ell_N} + e_{\ell_N}\}. \end{aligned} \quad (\text{AW23})$$

This last lower bound can be zero although the actual $\mathcal{B}(1)$ is strictly positive, as proved above. Every step concerns the original $q = 1$ quotient and its degree-two source. No extension of that dimension restriction to a quartet quotient is asserted.

3.1 Complete proof dependencies

The complete AT calculation accompanies this note, together with its 29 complete source dependencies and independent proof review. Those sources prove the original pushforward, tensor signs, full Taylor-unit injections, theta primitive, positive arithmetic and Gamma moments, and the quartet application. Every new assertion (AW7)–(AW23) is proved above on those unchanged finite objects. A separate independent review supplies the spectral calculation and its edge-case checks.

4 Joint trace control on the original source and relation spaces

Retain $h, k, \chi = \chi_{h,k}$ from the marked-product note. Here $q = \deg \chi \geq 1$ denotes the polynomial degree. Use the original coordinate $S = k/2 + iy$, the densities $m = w_h^{*k}$ and $r = r_{1/4}^{*k}$ with their literal masses, and $\mu_x = ((1-x)r + xm) dy$ for $0 \leq x \leq 1$. The common source is $H = \mathcal{P}_{2q}$ in the retained monomial basis. Its positive definite Gram and its original metric derivative are

$$M_x = M_0 + x\dot{M}, \quad M_0 = M_{2q}^\Gamma, \quad \dot{M} = M_{2q}^{\text{ar}} - M_{2q}^\Gamma, \quad T_x = M_x^{-1}\dot{M}. \quad (\text{SP1})$$

The equality $M_x T_x = \dot{M} = \dot{M}^*$ proves that T_x is self-adjoint in the M_x inner product. Its exact relation to the positive M_0 -self-adjoint operator $R = M_0^{-1} M_1$ is $T_x = (I + x(R - I))^{-1}(R - I)$, obtained by factoring $M_x = M_0(I + x(R - I))$. Thus its eigenvalues are $(r - 1)/(1 + x(r - 1))$ for the positive eigenvalues r of R , with their actual signs retained.

The precise arrow to the earlier AW source keeps H and every coefficient fixed, sends t_{AW} to x and u_{AW} to y , and sends the relation map by $B_N^{\text{SP}} = I_N B_N^{\text{AW}} : \mathcal{P}_{N-q} \rightarrow H$. The Gamma convolution constant is $c_a = 2^{1-2a} \Gamma(2a)$, so

$$r_{1/4}^{*k}(y) = \frac{c_{1/4}^k |\Gamma(k/4 + iy/2)|^2}{c_{k/4} 2\pi}, \quad c_{1/4} = \sqrt{2\pi}. \quad (\text{SP1a})$$

This is exactly AW3, including its mass $(2\pi)^{k/2}$; both arithmetic densities are w_h^{*k} with mass μ_h^k . Thus $M_x = M_{\text{AW}}(x)$, and $(B_N^{\text{SP}})^* M_x B_N^{\text{SP}} = (B_N^{\text{AW}})^* M_N B_N^{\text{AW}}$ with $M_N = I_N^* M_x I_N$.

4.1 All four endpoints as operators on one source

Let $I_N : \mathcal{P}_N \rightarrow H$ be literal coefficient inclusion, and let $B_N : \mathcal{P}_{N-q} \rightarrow H$ be multiplication by the original monic χ , followed by coefficient inclusion. Define

$$P_N = I_N(I_N^* M_x I_N)^{-1} I_N^* M_x, \quad Q_N = B_N(B_N^* M_x B_N)^{-1} B_N^* M_x, \quad Q_{q-1} = 0. \quad (\text{SP2})$$

The second formula is used for $N \geq q$; its Gram is invertible because multiplication by χ is injective on polynomials. Direct multiplication proves that these are the M_x -orthogonal projections onto the original source and relation subspaces. Within either family the subspaces are nested, so their orthogonal projections commute and their product is the smaller projection. Indeed the smaller subspace, its orthogonal complement inside the larger one, and the orthogonal complement of the larger one give three invariant summands on which this statement is immediate.

Let $V_N(x)$ be the original canonical quotient volume. In the fixed quotient-lift and monic-relation coordinates, block determinant gives

$$\log V_N(x) = \log \det(I_N^* M_x I_N) - \log \det(B_N^* M_x B_N). \quad (\text{SP3})$$

The relation determinant is one when its dimension is zero. The fixed monic change-of-basis determinant has modulus one, so SP3 retains every volume factor. Differentiation and cyclicity of finite matrix trace give

$$(\log V_N)' = \text{Tr}(T_x(P_N - Q_N)). \quad (\text{SP4})$$

For example $\text{Tr}(T_x P_N) = \text{Tr}((I_N^* M_x I_N)^{-1} I_N^* \dot{M} I_N)$; the relation formula uses precisely the same calculation. Set

$$F(x) = \log \frac{V_{q-1}(x) V_q(x)}{V_{2q-1}(x) V_{2q}(x)}, \quad \Delta_{h,k} = F(1) - F(0), \quad (\text{SP5})$$

and define two positive operators on this same source:

$$\mathcal{R}_x = Q_{2q-1} + Q_{2q} - Q_q, \quad \mathcal{P}_x = (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q), \quad \mathcal{A}_x = \mathcal{R}_x - \mathcal{P}_x. \quad (\text{SP6})$$

The four original signs in SP4 give

$$F'(x) = \text{Tr}(T_x \mathcal{A}_x). \quad (\text{SP7})$$

For the original row $v = v_{2q}(y)$, the density identity has the exact metric factor

$$R_x(y) - P_x(y) = v \mathcal{A}_x M_x^{-1} v^*.$$

Indeed $P_N M_x^{-1} = I_N(I_N^* M_x I_N)^{-1} I_N^*$, and the same identity holds with B_N, Q_N . This recovers the marked-product kernels and proves the common-source matrix comparison.

4.2 Ranks, multiplicities and the retained overlap

The following complete spectra and the final condition-number coefficient were already established in AW7 and AW14. They are retained here with the exact source correspondence above; the actual cross-trace and intersection terms supply the added refinement.

The nested relation spaces in SP6 have dimensions $1, q, q+1$. On the first space $\mathcal{R}_x$ has eigenvalue one; on the next orthogonal increment it has eigenvalue two with multiplicity $q-1$; on the last increment it has eigenvalue one; elsewhere it is zero. The nested source increments in $\mathcal{P}_x$ give the same list. When $q=1$, the multiplicity $q-1$ is zero. Consequently

$$\begin{aligned} \text{rank } \mathcal{R}_x &= \text{rank } \mathcal{P}_x = q+1, & \text{Tr } \mathcal{R}_x &= \text{Tr } \mathcal{P}_x = 2q, \\ \text{Tr } \mathcal{R}_x^2 &= \text{Tr } \mathcal{P}_x^2 = 4q-2, & \text{Tr } \mathcal{A}_x &= 0, \\ \text{Tr } \mathcal{A}_x^2 &= 8q-4-2\text{Tr}(\mathcal{R}_x \mathcal{P}_x). \end{aligned} \tag{SP8}$$

The equal kernel masses $2q$ are traces. The two positive operators have rank $q+1$, with their multiplicity-two directions retained.

The overlap in SP8 is at least one. Each positive operator is at least the orthogonal projection onto its range, since its nonzero eigenvalues are one and two. The two ranges each have dimension $q+1$ in dimension $2q+1$, so their intersection has dimension at least one. Choose a unit vector in this intersection and complete it to an orthonormal basis of the first range. The trace of the product of the two range projections is the sum of squared norms of the second projection on this basis, hence at least one. For positive operators, replacing one factor by a larger positive factor cannot decrease the trace of the product: the difference is $\text{Tr}(B^{1/2}(A' - A)B^{1/2}) \geq 0$. Applying this twice proves the stated bound for the actual operators.

4.3 A bound that uses cancellation before estimating

Write $d_x = \lambda_{\max}(T_x) - \lambda_{\min}(T_x)$. Since $\text{Tr } \mathcal{A}_x = 0$, for every real c

$$F'(x) = \text{Tr}((T_x - cI)\mathcal{A}_x). \tag{SP9}$$

Choose c to be the midpoint of the two extreme eigenvalues. In an M_x -orthonormal eigenbasis e_j of $\mathcal{A}_x$, with real eigenvalues α_j , one obtains

$$|F'(x)| \leq \sum_j |\alpha_j| |\langle e_j, (T_x - cI)e_j \rangle_{M_x}| \leq \frac{1}{2}d_x \sum_j |\alpha_j|.$$

This proves the trace-norm bound directly. Let L_x be the intersection of the two ranges in SP6, with $s_x = \dim L_x \geq 1$, and let E_{L_x} be its orthogonal projection in M_x . Each of $\mathcal{R}_x, \mathcal{P}_x$ is at least E_{L_x} . Thus $\mathcal{A}_x = (\mathcal{R}_x - E_{L_x}) - (\mathcal{P}_x - E_{L_x})$ is a difference of two positive operators of trace $2q - s_x$. The triangle inequality for trace norm gives $\|\mathcal{A}_x\|_1 \leq 4q - 2s_x \leq 4q - 2$. Cauchy-Schwarz on its at most $2q+1$ eigenvalues gives the refinement

$$|F'(x)| \leq \frac{d_x}{2} \min \left\{ 4q - 2s_x, \sqrt{(2q+1)(8q-4-2\text{Tr}(\mathcal{R}_x \mathcal{P}_x))} \right\}. \tag{SP10}$$

The overlap is calculated from the unchanged source and relation Grams in SP2. Every cross-pairing remains in that expression.

The complete earlier AW13-AW16 bound applies to this same source through $(U, W, C) = (\mathcal{R}_x, \mathcal{P}_x, T_x)$. Write $c_1(x) \leq \dots \leq c_{2q+1}(x)$ for all eigenvalues of T_x in M_x and define

$$S_{\text{AW}}(x) = c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \quad S_{\text{SP}}(x) = \text{the right-hand side of (SP10)}. \tag{SP10a}$$

Here is the complete retained spectral proof. For an orthogonal projection P of rank r , its diagonal values a_j in an orthonormal eigenbasis of T_x satisfy $0 \leq a_j \leq 1$ and $\sum_j a_j = r$. Thus $\text{Tr}(PT_x) = \sum_j c_j a_j$ is at least the sum of the smallest r and at most the sum of the largest r eigenvalues: moving a fixed diagonal mass from a smaller coefficient to a larger one increases this sum. Either $Z = \mathcal{R}_x$ or $Z = \mathcal{P}_x$ equals its range projection, of rank $q+1$, plus its eigenvalue-two projection, of rank $q-1$. Therefore

$$\sum_{j=1}^{q+1} c_j + \sum_{j=1}^{q-1} c_j \leq \text{Tr}(ZT_x) \leq \sum_{j=q+1}^{2q+1} c_j + \sum_{j=q+3}^{2q+1} c_j. \quad (\text{SP10b})$$

Subtracting the two extrema with their exact multiplicities gives $|F'(x)| \leq S_{\text{AW}}(x)$. Combining this bound with SP10 before integration proves

$$|F'(x)| \leq \min\{S_{\text{AW}}(x), S_{\text{SP}}(x)\}. \quad (\text{SP10c})$$

Positive smooth Gram matrices make the eigenvalues and overlaps bounded on $[0, 1]$; s_x is Borel since its rank strata are given by matrix minors. Hence the right side is integrable. The original source isometry of AW10 also yields the exact commutator identity $F' = \text{Tr}(W(T^{\text{iso}})^{-1}[C, T^{\text{iso}}])$. It follows from $U = T^{\text{iso}}W(T^{\text{iso}})^{-1}$ and cyclicity, with T^{iso} the explicit isometry and $C = T_x$ the metric derivative. Its full original observation remains $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q}$ and has exact defect $\mathcal{O}T^{\text{iso}} - \mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q}(T^{\text{iso}} - I)$ and kernel $\ker(\mathcal{O}T^{\text{iso}}) = (T^{\text{iso}})^{-1}(\chi\mathcal{P}_q)$. The defect follows by expansion and the kernel follows from injectivity of $\sigma_h^{\otimes k} \eta$ and $\ker J_{2q} = \chi\mathcal{P}_q$. Thus no additional cohomology-intertwining property of the metric isometry is used.

There is a closed integrated bound. Set $R = M_0^{-1}M_{2q}^{\text{ar}}$. It is positive and self-adjoint in the M_0 inner product, since $\langle v, Rv \rangle_{M_0} = \langle v, v \rangle_{M_{2q}^{\text{ar}}} > 0$. Let its smallest and largest eigenvalues be $r_{\min}, r_{\max} > 0$. The literal factorization of M_x gives

$$T_x = (I + x(R - I))^{-1}(R - I), \quad d_x = \frac{r_{\max} - 1}{1 + x(r_{\max} - 1)} - \frac{r_{\min} - 1}{1 + x(r_{\min} - 1)}. \quad (\text{SP11})$$

For fixed x , the function $(r - 1)/(1 + x(r - 1))$ is increasing on $r > 0$, its derivative being $(1 + x(r - 1))^{-2} > 0$. This proves the extreme-eigenvalue formula even though the inner product varies. Integration, including the identically zero integrand when an eigenvalue equals one, yields

$$\int_0^1 d_x dx = \log(r_{\max}/r_{\min}). \quad (\text{SP12})$$

Write $r_1 \leq \dots \leq r_{2q+1}$ for all positive eigenvalues of R , and retain

$$\mathcal{L}_q(R) = \log \frac{r_{q+2}}{r_q} + 2 \sum_{j=1}^{q-1} \log \frac{r_{2q+2-j}}{r_j}.$$

The ordered form of SP11 gives $c_j(x) = (r_j - 1)/(1 + x(r_j - 1))$ and $\int_0^1 c_j(x) dx = \log r_j$. Therefore SP7–SP12 and the full spectral proof above give both original-source controls

$$\boxed{|\Delta_{h,k}| \leq \int_0^1 \min\{S_{\text{AW}}(x), S_{\text{SP}}(x)\} dx \leq \mathcal{L}_q(R) \leq (2q - 1) \log \frac{r_{\max}}{r_{\min}},} \quad (\text{SP13})$$

$$|\Delta_{h,k}| \leq \frac{1}{2} \int_0^1 d_x \|\mathcal{R}_x - \mathcal{P}_x\|_1 dx \leq (2q - 1) \log \frac{r_{\max}}{r_{\min}}.$$

The first integral is also at most $\int_0^1 S_{\text{SP}}(x) dx$. Every logarithmic ratio in $\mathcal{L}_q$ is bounded by $\log(r_{\max}/r_{\min})$, with coefficient total $1 + 2(q - 1) = 2q - 1$; this proves the last inequality explicitly. The two displayed integral bounds each remain available: no ordering between the exact trace norm and the spectral refinement is assumed. When the Grams are proportional, every c_j agrees, both bounds are zero, and the correction is exactly zero without rescaling their original masses.

The exact trace norm and the centered Hilbert–Schmidt estimate are also retained before taking a minimum. On this same source define

$$\begin{aligned} S_{\text{tr}}(x) &= \frac{d_x}{2} \|\mathcal{R}_x - \mathcal{P}_x\|_1, \\ \Sigma(x) &= \text{Tr}(T_x^2) - \frac{(\text{Tr } T_x)^2}{2q+1}, \\ S_{\text{HS}}(x) &= \sqrt{\Sigma(x) \text{Tr}((\mathcal{R}_x - \mathcal{P}_x)^2)}. \end{aligned} \tag{SP13d}$$

The trace-norm inequality follows by subtracting the midpoint of the extreme eigenvalues from the metric derivative, since the trace of the signed projection difference is zero. In an orthonormal eigenbasis of that difference, every diagonal derivative entry is bounded in modulus by half the spectral diameter; summing with the absolute eigenvalues proves the assertion. For the other bound subtract the trace mean $\text{Tr}(T_x)I/(2q+1)$ instead. The square of its Hilbert–Schmidt norm in the unchanged source metric is exactly $\Sigma(x)$. Hilbert–Schmidt Cauchy–Schwarz with the signed difference proves that its trace pairing is bounded by $S_{\text{HS}}(x)$. Positivity of both factors under the square root follows from their self-adjointness. Thus both functions bound the same absolute signed derivative on general positive source forms; the angle term below still uses the specified moment measures. These are RMT10a–10b and ISM25–26 through the already proved original coefficient and projection maps.

On the original moment source (SP1a), the later angle calculation propagates further at this endpoint. With $a_q = 2q - 1$ define

$$\begin{aligned} A(x) &= a_q \sqrt{1 - e^{-F(x)/a_q}} d_x, \\ J^{(5)} &= \int_0^1 \min\{S_{\text{AW}}(x), S_{\text{tr}}(x), S_{\text{SP}}(x), S_{\text{HS}}(x), A(x)\} dx, \\ I &= \int_0^1 \min \left\{ d_x, \frac{S_{\text{AW}}(x)}{a_q \sqrt{1 - e^{-F(x)/a_q}}}, \frac{S_{\text{SP}}(x)}{a_q \sqrt{1 - e^{-F(x)/a_q}}}, \right. \\ &\quad \left. \frac{S_{\text{tr}}(x)}{a_q \sqrt{1 - e^{-F(x)/a_q}}}, \frac{S_{\text{HS}}(x)}{a_q \sqrt{1 - e^{-F(x)/a_q}}} \right\} dx. \end{aligned} \tag{SP13a}$$

The exact map to RMT is $a_{\text{RMT}} = a_q$, $F = \mathcal{B}$, $T_x = C_x$ and $(\mathcal{R}_x, \mathcal{P}_x) = (U_x, W_x)$, with the same original moment measures by SP1a. RMT11–RMT12 proves $F > 0$ and $|F'| \leq A$: its full two angle lists keep the designated zero angle, and vanishing F would imply $1 \perp \chi \mathcal{P}_q$, contradicting $\int |\chi|^2 d\mu_x > 0$ through $\chi^{\#k}(S) = \sum_j \bar{\chi}_j(k-S)^j$. Thus the two-bound proof above remains valid for general positive forms, while on these actual moment forms its displayed first bound in SP13 strengthens to

$$|\Delta_{h,k}| \leq J^{(5)} \leq \int_0^1 \min\{S_{\text{AW}}(x), S_{\text{SP}}(x)\} dx. \tag{SP13b}$$

For every integer $\ell_N \geq 0$ the constructed signed approximation is applied on these exact endpoint forms as well. Use $M_z = M_x|_{x=z}$, $C_z = T_x|_{x=z}$, $A_z = (\mathcal{R}_x - \mathcal{P}_x)|_{x=z}$, and $D = R$. The same-source RMT/ISM function $\mathcal{B}$ is exactly F in SP5. Write $D = (M_0)^{-1}M_1$, with its complete ordered positive eigenvalues $b_1, \dots, b_{2q+1}$, and let A_z be this same signed projection difference. Define

$$\begin{aligned} \alpha_z^\circ &= \frac{2(1-z) + z(b_1 + b_{2q+1})}{2}, \quad H_z^\circ = I - \frac{(1-z)I + zD}{\alpha_z^\circ}, \\ C_z^{[\ell_N]} &= (\alpha_z^\circ)^{-1} \sum_{r=0}^{\ell_N} (H_z^\circ)^r (D - I), \quad R_z^{[\ell_N]} = (H_z^\circ)^{\ell_N+1} C_z, \\ E_z^{[\ell_N]} &= R_z^{[\ell_N]} - \frac{\text{Tr } R_z^{[\ell_N]}}{2q+1} I. \end{aligned} \tag{SP13ea}$$

The symbols $\alpha_z^\circ, H_z^\circ$ preserve every original source parameter and relation map; ℓ_N is only the finite series truncation. Multiplication of the finite geometric sum gives $C_z - C_z^{[\ell_N]} = R_z^{[\ell_N]}$. All factors are real functions of the positive relative operator D , hence self-adjoint in the original $M_z = M_0((1 - z)I + zD)$ metric. Since $\text{Tr } A_z = 0$, the derivative of the same four-volume function is its signed center $\text{Tr}(C_z^{[\ell_N]} A_z)$ plus $\text{Tr}(E_z^{[\ell_N]} A_z)$. Consequently

$$\begin{aligned} j_{\ell_N} &= \int_0^1 \text{Tr}(C_z^{[\ell_N]} A_z) dz, \\ e_{\ell_N} &= \int_0^1 \sqrt{\text{Tr}((E_z^{[\ell_N]})^2) \text{Tr}(A_z^2)} dz, \\ j_{\ell_N} - e_{\ell_N} &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_{\ell_N} + e_{\ell_N}. \end{aligned} \tag{SP13eb}$$

For this block $\mathcal{B}$ denotes precisely the four-volume function identified above. The signed inequalities follow by Hilbert–Schmidt Cauchy–Schwarz in that unchanged metric and integration of the exact derivative. The full residual calculation ISM29–37, equivalently RMT19–21, uses $\vartheta = (b_{2q+1} - b_1)/(b_{2q+1} + b_1) < 1$ and $K_D = \sum_j (|b_j - 1|/\min\{1, b_j\})^2$ to prove

$$0 \leq e_{\ell_N} \leq \vartheta^{\ell_N+1} \sqrt{K_D(8q-6)} \longrightarrow 0. \tag{SP13ec}$$

Indeed every eigenvalue of H_z° has modulus at most ϑ , centering the residual subtracts a nonnegative square from its squared Hilbert–Schmidt norm, and $\text{Tr}(A_z^2) \leq 8q - 6$ by the full flag spectra and the common range. This proves the bound with all eigenvalue multiplicities retained. The displayed integrals are exact; a numerical evaluation must retain its own integration error. Intersecting this proved signed interval with the already proved absolute and nonlinear intervals supplies the additional endpoint entries below, for every specified ℓ_N .

For $H_0 = 2 \text{arccosh}(e^{F(0)/(2a_q)})$, the derivative $[2 \text{arccosh}(e^{B/(2a_q)})]' = 1/(a_q \sqrt{1 - e^{-B/a_q}})$ and its inverse $2a_q \log \cosh(z/2)$ give the actual endpoint interval

$$\begin{aligned} F(1) &\geq \max \left\{ 0, F(0) - J^{(5)}, 2a_q \log \cosh \frac{\max\{0, H_0 - I\}}{2}, F(0) + j_{\ell_N} - e_{\ell_N} \right\}, \\ F(1) &\leq \min \left\{ F(0) + J^{(5)}, 2a_q \log \cosh \frac{H_0 + I}{2}, F(0) + j_{\ell_N} + e_{\ell_N} \right\}, \quad I \leq \log(r_{\max}/r_{\min}). \end{aligned} \tag{SP13c}$$

The nonlinear branches follow by dividing all five derivative bounds by the same positive factor, integrating, and applying the increasing inverse. Intersecting with the constructed signed interval (SP13eb) gives the additional signed branches; compactness and $F > 0$ justify every denominator. It is precisely RMT13–RMT22 on the stated unchanged original source, including the full local signed construction (SP13ea)–(SP13ec).

SP1–SP13 give finite two-sided control on every original packet. They do not bound the displayed condition number or projection overlap along the growing arithmetic family. The next calculation can use these exact quantities, their original moment formulae, and the newly retained singular-boundary geometry; none has been replaced by an assumed asymptotic estimate or an RH verdict.

5 Complete periodized arithmetic source and finite-cutoff proofs

5.1 Periodized source control: complete comparison calculations

This companion retains the complete supplied periodized-source note, including P1–P64 and P26a–P26b, as its proof source. It supplies the source regularity, coordinate and image calculations used in reading those formulas, makes the finite Fourier projection part of the exact operator diagram, and retains the sharper one-sided error furnished by the omitted positive Gram. The adjoining density companion supplies the explicit definitions and full exponential-tail and completion proof. The original source, remainder basis, arithmetic unit, and relations are retained throughout.

The asymmetric cutoff calculation below is also the calculation in FC5–FC7 and the four-endpoint paragraph following FC13 of the adjoining periodized-curvature control bridge. PSA23–PSA25 record its source and determinant steps, with the later AW/SP spectral and actual-overlap control propagated through the same original and sampled coefficient maps, and PSA27–PSA29 record the same finite Fourier boundary diagram as FC19–FC20. These are audit derivations of the same mathematics, not additional results created by repeating the calculation.

Fix the original packet polynomial h , its full canceled zero orders, and $g = 2\xi$. Write $v_h = g/h$, with removable values understood, and

$$w_h(t) = \frac{1}{2\pi} |v_h(1/2 + it)|^2, \quad m_{h,k} = w_h^{*k}. \quad (\text{PSA1})$$

These definitions must be inserted before the first use of v_h, w_h in the supplied standalone note. They specify the original density; no rescaling of g, h , or the tensor-sum coordinate is performed. The full proof that these functions have the required exponential tails, including canceled-root values, is in the adjoining density companion.

5.1.1 The original coordinates and the complete source domain

Let $k \geq 1$ and let F be a finite linear combination of tensors of strong-Schwartz functions on $(0, \infty)$ and their logarithmic derivatives. Use $\mathcal{K} = \mathbb{C}$ for $k = 1$ and $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$ otherwise. In the ordered coordinates $(r, z_1, \dots, z_{k-1})$, put

$$y_i = r + z_i \ (i < k), \quad y_k = r, \quad x_i = e^{y_i}. \quad (\text{PSA2})$$

The derivative matrix from (r, z) to $(y_1, \dots, y_k)$ has first column all ones, last row $(1, 0, \dots, 0)$ and the identity in the remaining $k-1$ columns. Expansion along its last row gives determinant $(-1)^{k-1}$. Thus the positive density is $d^k x = e^{kr + \sum z_i} dr dz$, with the orientation sign recorded separately. The map and its inverse are

$$\begin{aligned} (\mathcal{U}_k F)(r, z) &= e^{(kr + \sum z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r), \\ (\mathcal{U}_k^{-1} \psi)(x) &= (x_1 \cdots x_k)^{-1/2} \psi(\log x_k, \log x_1 - \log x_k, \dots, \log x_{k-1} - \log x_k). \end{aligned} \quad (\text{PSA3})$$

Substitution into the integral proves equality of the two squared norms and proves that these maps are mutually inverse Hilbert isometries. With $D_i = -x_i \partial_{x_i}$ and $D^{(k)} = \sum_i D_i$, simultaneous change of all y_i is exactly change of r with every z_i fixed. Differentiating the displayed factor, without altering the coordinate, gives

$$\mathcal{U}_k D^{(k)} = (-\partial_r + k/2) \mathcal{U}_k, \quad \partial_r^j \mathcal{U}_k = (-1)^j \mathcal{U}_k (D^{(k)} - k/2)^j. \quad (\text{PSA4})$$

For every $a > 0$ and integer $j \geq 0$, strong-Schwartz estimates on each original tensor factor imply

$$I_{a,j} := \int_{\mathbb{R}} e^{2a|r|} \|\partial_r^j \mathcal{U}_k F(r)\|_{\mathcal{K}}^2 dr < \infty. \quad (\text{PSA5})$$

Indeed after (PSA3)–(PSA4) this is the original $d^k x$ integral of $|(D^{(k)} - k/2)^j F|^2$ multiplied by $\max\{x_k^{2a}, x_k^{-2a}\}$. The maximum is bounded by the sum of those two weights, and every term is a finite integral of a finite sum of products of original strong-Schwartz functions. This proves (PSA5) on the stated source without asserting it for arbitrary Hilbert inputs.

For completeness it also yields normal convergence of periodization. For any Hilbert-valued H^1 function f on $[r, r+1]$, select a point where its squared norm is no larger than its integral, use the fundamental theorem of calculus, and apply Cauchy–Schwarz. The result is $\|f(r)\|^2 \leq 2 \int_r^{r+1} (\|f(t)\|^2 + \|f'(t)\|^2) dt$. Consequently, for $\psi = \mathcal{U}_k F$,

$$\|\partial_r^j \psi(r)\|_{\mathcal{K}}^2 \leq 2e^{2a} e^{-2a|r|} (I_{a,j} + I_{a,j+1}). \quad (\text{PSA6})$$

The bound is summable over $r + mL$, uniformly for r in a fixed period and for every derivative. Thus the periodization is smooth; the ensuing Fourier transforms, products, and unfolded correlations are absolutely convergent on each finite source.

5.1.2 The full circle image and its Fourier constants

Use the plus-sign transform $\widehat{\psi}(u) = \int \psi(r) e^{iur} dr$ and the orthonormal circle basis $e_n(r) = L^{-1/2} e^{-2\pi i n r / L}$. Unfolding its coefficient gives

$$\langle e_n, \mathcal{P}_L \psi \rangle = L^{-1/2} \int_{\mathbb{R}} \psi(r) e^{2\pi i n r / L} dr, \quad \mathcal{P}_L \psi = \sum_m \psi(r + mL). \quad (\text{PSA7})$$

Here the scalar basis is paired with the vector-valued function using the conjugate-linear first-slot convention. In particular the zero coefficient is $L^{-1/2} \widehat{\psi}(0)$, whereas the constant circle function is $L^{-1} \widehat{\psi}(0)$.

The augmented coefficient map sends ψ to the nonzero coefficients and the unscaled vector $\widehat{\psi}(0)$. Its last summand has metric $L^{-1} \langle \cdot, \cdot \rangle_{\mathcal{K}}$. The inverse coordinate map on this image inserts $L^{-1/2} \widehat{\psi}(0)$ at $n = 0$. Thus, for a finite column map $\Psi = \mathcal{U}_k T$ and $Z_0 c = \widehat{\Psi} c(0)$,

$$M(L) = M^\times(L) + L^{-1} Z_0^* Z_0. \quad (\text{PSA8})$$

For $k = 1$, $T = \Theta \phi$, the Connes–Consani map remains $\mathcal{E} \phi = \frac{1}{2} \mathcal{U}_1 \Theta \phi$. It therefore has zero coefficient $\mathcal{M} \Theta \phi(1/2) / (2\sqrt{L})$. For the original seed this is $\xi(1/2) / \sqrt{L}$. Its entire Gram is one quarter of the original theta-source Gram. The coefficient and metric formulas record that factor explicitly.

The scalar convolution formula retains the relative Hilbert fibre. For $\psi = \mathcal{U}_k F_h^{\otimes k}$, take the further plus-sign Fourier transform in z . Its phase becomes

$$ur + \sum_{i < k} t_i z_i = \sum_{i < k} t_i y_i + \left(u - \sum_{i < k} t_i \right) y_k. \quad (\text{PSA9})$$

Fubini therefore gives the full amplitude $\prod_{i < k} v_h(1/2 + it_i) v_h(1/2 + i(u - \sum_{i < k} t_i))$. Plancherel in exactly the $k - 1$ relative variables gives

$$\begin{aligned} \|\widehat{\psi}(u)\|_{\mathcal{K}}^2 &= (2\pi)^{-(k-1)} \int_{\mathbb{R}^{k-1}} \prod_{i < k} |v_h(1/2 + it_i)|^2 |v_h(1/2 + i(u - \sum_{i < k} t_i))|^2 dt \\ &= 2\pi m_{h,k}(u). \end{aligned} \quad (\text{PSA10})$$

By (PSA4), a polynomial column contributes its factor $P(k/2 + iu)$. Parseval and (PSA7) now prove the entire sampled matrix

$$M_N(L)_{ab} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}(2\pi n/L) \overline{P_a(k/2 + 2\pi i n/L)} P_b(k/2 + 2\pi i n/L). \quad (\text{PSA11})$$

The density companion proves positive weights for $k \geq 2$ and hence strict positivity of this finite polynomial Gram for every $L > 0$. For $k = 1$, quantitative source comparison below supplies positivity once the stated period has been chosen; no assertion about positivity at every integer atom is substituted for that comparison.

5.1.3 The original-source correlation inequality

Define $M = \Psi^* \Psi$ and $M_{a,\pm} = (e^{\pm ar} \Psi)^* (e^{\pm ar} \Psi)$ in the unchanged coefficient basis. In original coordinates these are exactly the integrals weighted by $x_k^{\pm 2a}$. With $c^* \mathcal{C}(s) d = \int \langle \Psi c(r), \Psi d(r+s) \rangle dr$, unfolding pairs of translates gives

$$M(L) = \sum_{m \in \mathbb{Z}} \mathcal{C}(mL), \quad \mathcal{C}(0) = M, \quad \mathcal{C}(-s) = \mathcal{C}(s)^*. \quad (\text{PSA12})$$

For $s > 0$, insert e^{-ar} on the first factor and $e^{a(r+s)}$ on the second. Their product is e^{as} times the original integrand. Cauchy–Schwarz and the substitution $t = r + s$ give

$$|c^* \mathcal{C}(s) c| \leq e^{-as} (c^* M_{a,-} c)^{1/2} (c^* M_{a,+} c)^{1/2}. \quad (\text{PSA13})$$

The roles of the weights reverse for negative s . The two geometric series sum to $2/(e^{aL}-1)$. Applying $2\sqrt{xy} \leq x+y$ to these actual two quadratic forms yields

$$-\frac{M_{a,+} + M_{a,-}}{e^{aL} - 1} \preceq M(L) - M \preceq \frac{M_{a,+} + M_{a,-}}{e^{aL} - 1}. \quad (\text{PSA14})$$

This proves an inequality on every coefficient vector, including complex cross-pairings, and hence the claimed matrix order.

5.1.4 All-period recovery before the quotient minimum

For $\delta > 0$, quadratic-form Tonelli and $u = 2\pi n/L$ for $n > 0$ give

$$\int_0^\infty L^{-2-\delta} \|\widehat{\Psi}_C(2\pi n/L)\|^2 dL = (2\pi n)^{-1-\delta} \int_0^\infty u^\delta \|\widehat{\Psi}_C(u)\|^2 du. \quad (\text{PSA15})$$

Negative n supplies the negative half-line. The sum of $n^{-1-\delta}$ over the positive integers is exactly $\zeta(1+\delta)$, with no double counting of either half-line. Equality of quadratic forms proves

$$M_\delta := \frac{1}{2\pi} \int |u|^\delta \widehat{\Psi}(u)^* \widehat{\Psi}(u) du = \frac{(2\pi)^\delta}{\zeta(1+\delta)} \int_0^\infty L^{-1-\delta} M^\times(L) dL. \quad (\text{PSA16})$$

For $0 < \delta \leq \delta_0$, the mean-value formula implies $||u|^\delta - 1| \leq \delta |\log |u|| (1 + |u|^{\delta_0})$. The right side is integrable against every finite-source Fourier Gram: the logarithm is locally integrable and the tails are Schwartz. Thus the exact matrix with this logarithmic integrand, denoted $\mathcal{L}_{\delta_0}$ in P31, obeys $-\delta \mathcal{L}_{\delta_0} \preceq M_\delta - M \preceq \delta \mathcal{L}_{\delta_0}$, proving convergence and its rate. If the nonzero $Z_0^* Z_0/L$ term were inserted into (PSA16), its small-period integral would be $\int_0^\infty L^{-2-\delta} dL$ times that matrix and would diverge. Formula (PSA8) is the exact reconstruction of the omitted coefficient. Formula (PSA16) averages source Grams; it makes no interchange of this integral with inverse compression.

5.1.5 The same finite quotient and its actual relation coefficient

Let $E = \mathbb{C}[S]/(\chi)$, $q = \deg \chi > 0$, and $\mathcal{P}_N = \{P : \deg P \leq N\}$, with its original basis. For $N \geq q-1$, the monic remainder map $J_N : \mathcal{P}_N \rightarrow E$ is onto. The actual arithmetic observation in the larger tensor target is still ηJ_N , with $\eta[P] = v_h^{\otimes k} P(A_k)1$ and $v_h = j_h(g/h)$. We take determinants only in the stated remainder basis of E .

For any positive source Gram M_N , set

$$K_N = J_N M_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad C_N = M_N^{-1} J_N^* G_N. \quad (\text{PSA17})$$

Surjectivity makes K_N positive definite. Direct multiplication gives $J_N C_N = I$. If $J_N P = v$, then $P = C_N v + z$ with $J_N z = 0$. The cross-term vanishes because $C_N^* M_N z = G_N J_N z = 0$. Therefore

$$P^* M_N P = v^* G_N v + z^* M_N z, \quad v^* G_N v = \min_{J_N P = v} P^* M_N P. \quad (\text{PSA18})$$

This proves both the actual minimum and its unique representative. Applying any source lower and upper bounds to this same affine set and then taking its minimum transfers those same factors to G_N .

In particular, (PSA14) and the actual Rayleigh quotient $\kappa_{a,N} = \max_{c \neq 0} c^*(M_{a,+,N} + M_{a,-,N})c / (c^* M_N c)$ prove P34–P37. The constraint set and coefficient coordinates are unchanged; the minimizing representatives may change and their difference is retained below. For the second representative $C_N(L)$ obtained from $M_N(L)$,

$$J_N(C_N(L) - C_N) = 0, \quad C_N(L) - C_N = M_\chi \operatorname{div}_\chi(C_N(L) - C_N). \quad (\text{PSA19})$$

Monic division here is applied column by column and yields the actual degree- $\leq N - q$ coefficient, with the zero coefficient if $N = q - 1$. The tensor relation is also explicit. In the original polynomial ring, fixed-order monic division in the variables $s_1, \dots, s_k$ expresses $\chi(\sum s_i)Q(\sum s_i)$ as $\sum_i h(s_i)Q_i$. Its remainder is zero because χ is the annihilator of the original tensor-sum cyclic action. Replace the i th F_h by ϕ_* , apply $Q_i(D_1, \dots, D_k)$, and attach $(-1)^{i-1}$. The tensor differential supplies the other $(-1)^{i-1}$; their product is $+1$, and $h(D_i)F_h = \Theta\phi_*$ supplies the original relation. Its arithmetic image is the receiving supported zero, with its source coefficient retained, and not external absence.

5.1.6 The finite frequency cutoff and its one-sided allowance

For $p \geq 1$, define the unchanged-source derivative Gram

$$H_{p,N} = \int (1 + r^2)(\partial_r^p \Psi_N)^* \partial_r^p \Psi_N dr. \quad (\text{PSA20})$$

By (PSA4), the original-coordinate source map is $(-1)^p(D^{(k)} - k/2)^p$ and its weight is $1 + (\log x_k)^2$. Integration by parts has no boundary term by (PSA6), and gives $\widehat{\partial_r^p \psi}(u) = (-iu)^p \widehat{\psi}(u)$. Weighted Cauchy-Schwarz with $\int (1 + r^2)^{-1} dr = \pi$ proves $\|\widehat{\Psi_N^c}(u)\|^2 \leq \pi |u|^{-2p} c^* H_{p,N} c$ for $u \neq 0$.

Let $\Pi_{L,J}$ be the orthogonal projection onto precisely the Fourier modes $|n| \leq J$, including zero, and form $M_N(L, J) = (\Pi_{L,J} \mathcal{P}_L \Psi_N)^* (\Pi_{L,J} \mathcal{P}_L \Psi_N)$. The omitted Gram is positive. Summing both signs of n and using $\sum_{n>J} n^{-2p} \leq \int_J^\infty t^{-2p} dt$ proves

$$0 \leq M_N(L) - M_N(L, J) \leq \frac{1}{2p-1} \left(\frac{L}{2\pi J} \right)^{2p-1} H_{p,N}. \quad (\text{PSA21})$$

The exact coefficient before the series was $(2\pi/L)(L/(2\pi))^{2p}$; (PSA21) retains the two-sided lattice factor and the original Plancherel convention.

Use the actual polynomial inclusion from degree N into degree D . It pulls back each of $M_D, M_{a,\pm,D}, H_{p,D}$ to the corresponding degree- N form. Consequently their Rayleigh bounds at D control all $N \leq D$. With the exact P62 quantities, write

$$\eta_{\text{per}} = \frac{\kappa_{a,D}}{e^{aL} - 1}, \quad \eta_{\text{tail}} = \frac{\lambda_{p,D}}{2p-1} \left(\frac{L}{2\pi J} \right)^{2p-1}, \quad \lambda_{p,D} = \max_{c \neq 0} \frac{c^* H_{p,D} c}{c^* M_D c}. \quad (\text{PSA22})$$

Choose L and integer J by P64 for specified positive budgets whose sum is below one. These choices prove that $\eta_{\text{per}} + \eta_{\text{tail}} < 1$ after choosing the budgets strictly below one. Combining the two actual inequalities gives the sharper, asymmetric result

$$(1 - \eta_{\text{per}} - \eta_{\text{tail}})M_N \preceq M_N(L, J) \preceq (1 + \eta_{\text{per}})M_N. \quad (\text{PSA23})$$

The upper bound retains the positivity of the omitted Gram. It implies the supplied symmetric total-error bound, and also proves strict positivity of the finite sampled matrix. For $k \geq 2, J \geq D$ in P64 separately supplies more distinct positive sample atoms than the degree of a nonzero polynomial.

Define $G_N(L, J), C_N(L, J)$ by (PSA17) using this finite Gram. By (PSA18), the factors in (PSA23) pass unchanged to $G_N(L, J)$. The same proof of (PSA19) supplies its original theta correction. Taking determinants on the fixed q -dimensional remainder space gives

$$q \log(1 - \eta_{\text{per}} - \eta_{\text{tail}}) \leq \log \frac{\det G_N(L, J)}{\det G_N} \leq q \log(1 + \eta_{\text{per}}). \quad (\text{PSA24})$$

For $D = 2q + 2$, all four endpoint degrees and both generator raises remain among these actual restrictions. The common space for the four determinant terms themselves is $\mathcal{H} = \mathcal{P}_{2q}$. Let $I_N : \mathcal{P}_N \hookrightarrow \mathcal{H}$ and $I_{2q,D} : \mathcal{H} \hookrightarrow \mathcal{P}_D$ be the literal coefficient inclusions. Restriction gives

$$M^{(0)} = I_{2q,D}^* M_D I_{2q,D} = M_{2q}, \quad M^{(1)} = I_{2q,D}^* M_D(L, J) I_{2q,D} = M_{2q}(L, J). \quad (\text{PSA24a})$$

Set $\alpha = 1 - \eta_{\text{per}} - \eta_{\text{tail}} > 0$ and $\beta = 1 + \eta_{\text{per}}$. These two forms satisfy $\alpha M^{(0)} \preceq M^{(1)} \preceq \beta M^{(0)}$ by (PSA23). In this paragraph let $x \in [0, 1]$ be a metric path parameter, distinct from the physical x_i , and retain

$$M(x) = (1 - x)M^{(0)} + xM^{(1)}, \quad D_{L,J} = (M^{(0)})^{-1}M^{(1)}, \quad C(x) = M(x)^{-1}(M^{(1)} - M^{(0)}). \quad (\text{PSA24b})$$

These are the original and actual sampled forms, rather than a Gamma comparison. The finite algebra of the AW and SP calculations applies through the identity of this coefficient space, with their endpoint forms replaced by exactly $M^{(0)}, M^{(1)}$. We prove that algebra for this source here.

For $N \in \{q - 1, q, 2q - 1, 2q\}$ put $B_N = \times \chi : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_N$ and define

$$\begin{aligned} P_N(x) &= I_N(I_N^* M(x) I_N)^{-1} I_N^* M(x), \\ Q_N(x) &= I_N B_N (B_N^* I_N^* M(x) I_N B_N)^{-1} B_N^* I_N^* M(x), \\ U(x) &= Q_{2q-1} + Q_{2q} - Q_q, \quad Q_{q-1} = 0, \\ W(x) &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q). \end{aligned} \quad (\text{PSA24c})$$

Every inverse is on the displayed actual domain. For $N = q - 1$ the relation domain is zero and its determinant is one. Multiplication verifies that these are the orthogonal projections onto the specified polynomial and relation subspaces in $M(x)$. The degree-less-than- q remainder section s_N and the monic relation columns give a frame $[s_N, B_N]$ of determinant one: its high-degree block is triangular with diagonal one. Block elimination in its Gram therefore gives

$$V_N(x) = \frac{\det(I_N^* M(x) I_N)}{\det(B_N^* I_N^* M(x) I_N B_N)}, \quad \frac{d}{dx} \log V_N(x) = \text{Tr}((P_N - Q_N)C(x)). \quad (\text{PSA24d})$$

This determinant is the same quotient determinant as (PSA17)–(PSA18), since its Schur complement is the minimum Gram over the same remainder fibres. The derivative follows by the finite determinant derivative and cyclicity of trace. In the relation-first orientation, the unchanged frame has determinant $(-1)^{q(N-q+1)}$ and its squared modulus is one, so no orientation factor has been removed. Consequently the four original signs give $\mathcal{B}'(x) = \text{Tr}((U - W)C(x))$.

The flags of relation subspaces have dimensions $1, q, q + 1$, and the polynomial flag has dimensions $q, q + 1, 2q, 2q + 1$. Nested projections act as identity or zero on successive orthogonal pieces. Hence each of U, W has eigenvalues 0 of multiplicity q , 1 of multiplicity 2, and 2 of multiplicity $q - 1$. Thus each has rank $q + 1$, trace $2q$ and trace of the square $\text{Tr } U^2 = \text{Tr } W^2 = 4q - 2$. The eigenvalue-two multiplicity is zero at $q = 1$. Let $s(x) = \dim(\text{ran } U \cap \text{ran } W) \geq 1$, and write $c_1(x) \leq \dots \leq c_n(x)$ for the eigenvalues of $C(x)$, $n = 2q + 1$, and $d(x) = c_n(x) - c_1(x)$. Define

$$\begin{aligned} S_{\text{AW}}^{L,J}(x) &= c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \\ S_{\text{SP}}^{L,J}(x) &= \frac{d(x)}{2} \min \left\{ 4q - 2s(x), \sqrt{(2q+1)(8q-4-2\text{Tr}(UW))} \right\}, \\ \mathcal{J}_{L,J} &= \int_0^1 \min \{ S_{\text{AW}}^{L,J}(x), S_{\text{SP}}^{L,J}(x) \} dx. \end{aligned} \quad (\text{PSA24e})$$

For an orthogonal projection P of rank r , its diagonal entries in a $C(x)$ -eigenbasis lie in $[0, 1]$ and sum to r . Moving a fixed diagonal mass from a smaller eigenvalue to a larger one increases its trace pairing; hence $\sum_{j=1}^r c_j \leq \text{Tr}(PC) \leq \sum_{j=n-r+1}^n c_j$. Each of U, W is the sum of a rank- $q + 1$ range projection and a rank- $q - 1$ eigenvalue-two projection. Applying these bounds and subtracting their full extrema yields $|\text{Tr}((U - W)C)| \leq S_{\text{AW}}^{L,J}$.

Let E_L be the projection onto the actual intersection of the two ranges. The positive operators $U - E_L, W - E_L$ both have trace $2q - s(x)$, because each of U, W dominates its range projection. Thus $\|U - W\|_1 \leq 4q - 2s(x)$. Also $\text{Tr}(U - W) = 0$ and $\text{Tr}((U - W)^2) = 8q - 4 - 2\text{Tr}(UW)$. Cauchy-Schwarz on its n real eigenvalues gives the other trace-norm estimate in (PSA24e). Subtracting $(c_1 +$

$c_n)I/2$ from C leaves the trace pairing unchanged and bounds its absolute value by $d(x)\|U - W\|_1/2$, as is seen in an orthonormal eigenbasis of $U - W$. This proves the SP bound on the same signed derivative, and therefore its pointwise minimum with the AW bound. Smooth positive forms give bounded projector matrices and continuous eigenvalues; rank strata are defined by matrix minors, so $s(x)$ is Borel. The minimum is integrable, and integration with its original signs gives the first bound below.

The exact trace norm and the centered Hilbert–Schmidt estimate are also retained before taking a minimum. On this same source define

$$\begin{aligned} S_{\text{tr}}(x) &= \frac{d(x)}{2}\|U - W\|_1, \\ \Sigma(x) &= \text{Tr}(C(x)^2) - \frac{(\text{Tr } C(x))^2}{2q+1}, \\ S_{\text{HS}}(x) &= \sqrt{\Sigma(x) \text{Tr}((U - W)^2)}. \end{aligned} \tag{PSA24g}$$

The trace-norm inequality follows by subtracting the midpoint of the extreme eigenvalues from the metric derivative, since the trace of the signed projection difference is zero. In an orthonormal eigenbasis of that difference, every diagonal derivative entry is bounded in modulus by half the spectral diameter; summing with the absolute eigenvalues proves the assertion. For the other bound subtract the trace mean $\text{Tr}(C(x))I/(2q+1)$ instead. The square of its Hilbert–Schmidt norm in the unchanged source metric is exactly $\Sigma(x)$. Hilbert–Schmidt Cauchy–Schwarz with the signed difference proves that its trace pairing is bounded by $S_{\text{HS}}(x)$. Positivity of both factors under the square root follows from their self-adjointness. Thus both functions bound the same absolute signed derivative on general positive source forms; the angle term below still uses the specified moment measures. These are RMT10a–10b and ISM25–26 through the already proved original coefficient and projection maps.

The actual source in (PSA24a) has additional moment structure. The first measure is $m_{h,k}(y) dy$, and the second is the unchanged positive atomic measure $\sum_{|n| \leq J} (2\pi/L) m_{h,k}(2\pi n/L) \delta_{2\pi n/L}$, including zero. Their degree- $2q$ Grams are exactly $M^{(0)}, M^{(1)}$ by (PSA10)–(PSA11) and (PSA21). Positivity was proved in (PSA23). Hence the complete RMT calculation applies with this original/sampled pair, with no change to any atom, mass or polynomial coefficient. For $a_q = 2q - 1$ put

$$\begin{aligned} A_{L,J}(x) &= a_q \sqrt{1 - e^{-\mathcal{B}(x)/a_q}} d(x), \\ \mathcal{J}_{L,J}^{(5)} &= \int_0^1 \min\{S_{\text{AW}}^{L,J}(x), S_{\text{tr}}(x), S_{\text{SP}}^{L,J}(x), S_{\text{HS}}(x), A_{L,J}(x)\} dx, \\ \mathcal{I}_{L,J} &= \int_0^1 \min \left\{ d(x), \frac{S_{\text{AW}}^{L,J}(x)}{a_q \sqrt{1 - e^{-\mathcal{B}(x)/a_q}}}, \frac{S_{\text{SP}}^{L,J}(x)}{a_q \sqrt{1 - e^{-\mathcal{B}(x)/a_q}}}, \right. \\ &\quad \left. \frac{S_{\text{tr}}(x)}{a_q \sqrt{1 - e^{-\mathcal{B}(x)/a_q}}}, \frac{S_{\text{HS}}(x)}{a_q \sqrt{1 - e^{-\mathcal{B}(x)/a_q}}} \right\} dx. \end{aligned} \tag{PSA24f}$$

RMT11 keeps both original q -angle lists of the crossed quotient pairs and proves $|\mathcal{B}'| \leq A_{L,J}$ on these measures. Its strict positivity argument applies also to this atomic endpoint: equality $\mathcal{B} = 0$ would place 1 orthogonal to $\chi \mathcal{P}_q$, and pairing with $\chi^{\#k}(S) = \sum_j \overline{\chi_j}(k - S)^j$ would give $0 = \int |\chi|^2 d\mu_x$, contradicting the positive Gram of the nonzero χ . Thus $\mathcal{B} > 0$ on the whole compact path. Taking the minimum of this angle derivative bound, the two earlier fully proved bounds, and both bounds in (PSA24g) yields $|\mathcal{B}(1) - \mathcal{B}(0)| \leq \mathcal{J}_{L,J}^{(5)} \leq \mathcal{I}_{L,J}$. The generic two-form proof remains valid; this further inequality uses precisely the actual moment measures.

Let $b_1 \leq \dots \leq b_n$ be all eigenvalues of $D_{L,J}$. The form comparison gives $\alpha \leq b_j \leq \beta$. Factoring $M(x) = M^{(0)}((1-x)I + xD_{L,J})$ proves $c_j(x) = (b_j - 1)/(1 - x + xb_j)$; the order follows from derivative $(1 - x + xb_j)^{-2} > 0$ with respect to b_j . Its integral is $\log b_j$. Therefore the full strengthened

finite-circle error, with every original source constant, is

$$\begin{aligned}
|\mathcal{B}_{h,k}(L, J) - \mathcal{B}_{h,k}| &\leq \mathcal{J}_{L,J}^{(5)} \leq \mathcal{J}_{L,J} \\
&\leq \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \\
&\leq (2q-1) \log \frac{1 + \eta_{\text{per}}}{1 - \eta_{\text{per}} - \eta_{\text{tail}}}.
\end{aligned} \tag{PSA25}$$

The first quantity is also at most $\int_0^1 S_{\text{SP}}^{L,J}(x) dx$. Each logarithmic ratio in the middle expression is at most $\log(\beta/\alpha)$ and its coefficient total is $2q-1$. This proves the last inequality, including the empty sum at $q=1$. It is at most $(2q-1) \log((1+\eta)/(1-\eta))$ for $\eta = \eta_{\text{per}} + \eta_{\text{tail}}$. The older bound with coefficient $2q$ follows from this stronger estimate; the asymmetric omitted-tail allowance is unchanged. Fixed positive budgets still give an $O(q_k)$ error, whose ratio to $q_k \log k$ tends to zero. No bound on the growth of the original $\kappa_{a,D}$ or $\lambda_{p,D}$ is inserted. The sampled values remain the exact analytic values with every zero mode, Fourier factor and tensor source coefficient retained.

For every integer $\ell_N \geq 0$ the constructed signed approximation is applied on these exact end-point forms as well. Use $M_z = M(x)|_{x=z}$, $C_z = C(x)|_{x=z}$, $A_z = (U(x) - W(x))|_{x=z}$ and $D = D_{L,J}$. These are the actual continuous/sampled moment forms in (PSA24a), and the RMT/ISM function $\mathcal{B}$ is precisely the original sampled-path four-volume function. The truncation ℓ_N differs from both the circle length L and the original derivative order p . Write $D = (M_0)^{-1}M_1$, with its complete ordered positive eigenvalues $b_1, \dots, b_{2q+1}$, and let A_z be this same signed projection difference. Define

$$\begin{aligned}
\alpha_z^\circ &= \frac{2(1-z) + z(b_1 + b_{2q+1})}{2}, \quad H_z^\circ = I - \frac{(1-z)I + zD}{\alpha_z^\circ}, \\
C_z^{[\ell_N]} &= (\alpha_z^\circ)^{-1} \sum_{r=0}^{\ell_N} (H_z^\circ)^r (D - I), \quad R_z^{[\ell_N]} = (H_z^\circ)^{\ell_N+1} C_z, \\
E_z^{[\ell_N]} &= R_z^{[\ell_N]} - \frac{\text{Tr } R_z^{[\ell_N]}}{2q+1} I.
\end{aligned} \tag{PSA25ba}$$

The symbols $\alpha_z^\circ, H_z^\circ$ preserve every original source parameter and relation map; ℓ_N is only the finite series truncation. Multiplication of the finite geometric sum gives $C_z - C_z^{[\ell_N]} = R_z^{[\ell_N]}$. All factors are real functions of the positive relative operator D , hence self-adjoint in the original $M_z = M_0((1-z)I + zD)$ metric. Since $\text{Tr } A_z = 0$, the derivative of the same four-volume function is its signed center $\text{Tr}(C_z^{[\ell_N]} A_z)$ plus $\text{Tr}(E_z^{[\ell_N]} A_z)$. Consequently

$$\begin{aligned}
j_{\ell_N} &= \int_0^1 \text{Tr}(C_z^{[\ell_N]} A_z) dz, \\
e_{\ell_N} &= \int_0^1 \sqrt{\text{Tr}((E_z^{[\ell_N]})^2) \text{Tr}(A_z^2)} dz, \\
j_{\ell_N} - e_{\ell_N} &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_{\ell_N} + e_{\ell_N}.
\end{aligned} \tag{PSA25bb}$$

For this block $\mathcal{B}$ denotes precisely the four-volume function identified above. The signed inequalities follow by Hilbert–Schmidt Cauchy–Schwarz in that unchanged metric and integration of the exact derivative. The full residual calculation ISM29–37, equivalently RMT19–21, uses $\vartheta = (b_{2q+1} - b_1)/(b_{2q+1} + b_1) < 1$ and $K_D = \sum_j (|b_j - 1|/\min\{1, b_j\})^2$ to prove

$$0 \leq e_{\ell_N} \leq \vartheta^{\ell_N+1} \sqrt{K_D(8q-6)} \longrightarrow 0. \tag{PSA25bc}$$

Indeed every eigenvalue of H_z° has modulus at most ϑ , centering the residual subtracts a nonnegative square from its squared Hilbert–Schmidt norm, and $\text{Tr}(A_z^2) \leq 8q-6$ by the full flag spectra

and the common range. This proves the bound with all eigenvalue multiplicities retained. The displayed integrals are exact; a numerical evaluation must retain its own integration error. Intersecting this proved signed interval with the already proved absolute and nonlinear intervals supplies the additional endpoint entries below, for every specified ℓ_N .

This also improves the actual sampled endpoint itself in its nonlinear coordinate. Define $H_0 = 2 \operatorname{arcosh}(e^{\mathcal{B}_{h,k}/(2a_q)})$. RMT16's derivative and inverse, on this identified moment path, give $|\mathcal{H}_{a_q}(\mathcal{B}_{h,k}(L, J)) - H_0| \leq \mathcal{I}_{L,J} \leq \log(b_n/b_1) \leq \log(\beta/\alpha)$. Combining the increasing inverse with the just-proved additive bound yields

$$\begin{aligned} \mathcal{B}_{h,k}(L, J) &\geq \max \left\{ 0, \mathcal{B}_{h,k} - \mathcal{J}_{L,J}^{(5)}, 2a_q \log \cosh \frac{\max\{0, H_0 - \mathcal{I}_{L,J}\}}{2}, \mathcal{B}_{h,k} + j_{\ell_N} - e_{\ell_N} \right\}, \\ \mathcal{B}_{h,k}(L, J) &\leq \min \left\{ \mathcal{B}_{h,k} + \mathcal{J}_{L,J}^{(5)}, 2a_q \log \cosh \frac{H_0 + \mathcal{I}_{L,J}}{2}, \mathcal{B}_{h,k} + j_{\ell_N} + e_{\ell_N} \right\}. \end{aligned} \quad (\text{PSA25a})$$

All denominators in (PSA24f) are justified by the strict positivity above, with no uniform lower bound in tensor order presumed.

5.1.7 The finite observation map and both Laplacian boundary terms

Put $\Phi_{N,L,J} = \Pi_{L,J} \mathcal{P}_L \mathcal{U}_k \mathcal{V}_{h,k}|_{\mathcal{P}_N}$. Strict positivity in (PSA23) makes it injective. The exact observation has the stated image as domain:

$$\mathcal{J}_{N,L,J} : \operatorname{im} \Phi_{N,L,J} \longrightarrow E, \quad \mathcal{J}_{N,L,J} = J_N \Phi_{N,L,J}^{-1}. \quad (\text{PSA26})$$

On that image, the inverse can equivalently be written $(\Phi_{N,L,J}^* \Phi_{N,L,J})^{-1} \Phi_{N,L,J}^*$ in the unchanged coefficient coordinates. Its kernel is $\Phi_{N,L,J}(\ker J_N)$, so its quotient is exactly E with the metric (PSA17), retaining all jets. This construction uses finite dimensionality and the proved lower source bound; it asserts no bounded inverse on a larger completed source.

On the periodic H^1 domain let $D_L = -\partial_r + k/2$. The adjoint is $D_L^* = \partial_r + k/2$, so $D_L^* + D_L = k$. On periodic H^2 , $\mathcal{L}_{L,k} = D_L^2 - kD_L = \partial_r^2 - k^2/4$ is the self-adjoint Fourier diagonal with values $-(2\pi n/L)^2 - k^2/4$. The projection $\Pi_{L,J}$ commutes with both derivatives on these domains.

Let $r = \mathcal{V}_{h,k} C_N(L, J)$ and use the original arithmetic action $A = M_S$ on E . The finite representative and its actual boundary are

$$R = \Pi_{L,J} \mathcal{P}_L \mathcal{U}_k r, \quad B = \Pi_{L,J} \mathcal{P}_L \mathcal{U}_k (D^{(k)} r - rA). \quad (\text{PSA27})$$

The unperiodized difference has zero remainder at degree $N + 1$ and has the monic theta primitive already proved above. Commutation and (PSA4) prove $D_L R - RA = B$ and $R^* R = G_N(L, J)$. Substituting $RA = D_L R - B$ and using $D_L^* + D_L = k$ gives

$$A^* G_N(L, J) + G_N(L, J) A - k G_N(L, J) = -(R^* B + B^* R). \quad (\text{PSA28})$$

Applying D_L once more to the first identity and subtracting k times it gives, with every term retained,

$$\mathcal{L}_{L,k} R - R(A^2 - kA) = (D_L - k)B + BA. \quad (\text{PSA29})$$

This proves the finite-cutoff extension of P46–P48. Setting $\Pi_{L,J} = I$ gives the original full-period identities.

For $Av = \rho v$, (PSA29) is $(\mathcal{L}_{L,k} - \rho(\rho - k))Rv = ((D_L - k)B + BA)v$. On the original Fourier diagonal, the squared norm on the left is the sum of $|-(2\pi n/L)^2 - k^2/4 - \rho(\rho - k)|^2 \|(Rv)_n\|^2$. Every coefficient is at least $|\operatorname{Im}(\rho(\rho - k))|^2$, proving P49 also after the finite projection. The conclusion is a lower bound on this actual boundary, not an upper bound on the arithmetic endpoint volume. The exact observation (PSA26), its one- and two-degree extensions, and (PSA27)–(PSA29) supply the map from the circle calculation to the arithmetic action. For those inverse observation extensions, take the common controlled source degree $D \geq N + 2$; the endpoint choice $D = 2q + 2$ already has this property for every endpoint $N \leq 2q$. The boundary identity itself uses the displayed smooth columns and needs injectivity only at its representative degree N .

5.1.8 The subsequent completion and the scope of verification

The fixed-circle limit keeps L fixed and completes the increasing polynomial source. The adjoining density proof supplies its typed isometry to $\ell^2(\mathbb{Z}, \nu)$, polynomial density for the original measure and its $|\chi|^2$ weight, and the exact quotient map $[P] \mapsto (P(S_n))_{\chi(S_n)=0}$. Its kernel is $(p_L)/(\chi)$, with p_L the squarefree product of the sampled roots, and the canonical Grams decrease to the sampled-value Gram. That proof retains all multiplicities in χ before taking the calculated kernel. The finite observation (PSA26) instead retains the entire remainder space at the specified N, L, J . These are related by the explicit quotient and completion arrows; an unsampled class or a nonconstant sampled jet maps to a supported zero under the latter arrow.

The complete supplied 16-method finite suite passed both normally and under Python optimization, with identical result records. Seven additional in-memory mutations each failed at one intended mathematical assertion in both modes, with no execution errors. These finite checks cover stated algebraic interfaces. The original zero-mode command-line negative is an unconditional failure-path control; the additional mutation removes the actual zero-mode Gram from its original complex-matrix calculation. The original raw checker and proof sources remain unchanged. The analytic proofs above and the adjoining density proof remain written mathematics; no Lean or interval certificate is inferred from those checks.

6 Finite-circle curvature and complete quotient compensation

6.1 Finite sampling, residue curvature and the original control form

We retain the polynomial source, remainder map, coefficient basis and arithmetic inner product from the periodized-source construction. The following calculation propagates its proved finite-source errors to the residue insertion and the original Laplacian. The quotient curvature calculation also incorporates the exact Schur-complement argument supplied by the coordinating Zeta task. Fix the original tensor degree $k \in \mathbb{Z}_{\geq 1}$ throughout the source construction.

Let $E = \mathbb{C}[S]/(\chi)$, with the original monic χ of degree $q \geq 1$, $A = M_S$, $e_0 = [1]$, $\ell[P] = [S^{q-1}] \text{rem}_\chi P$, and $R = e_0 \ell$. For $N \geq q - 1$, let $\mathcal{P}_N$ be the original polynomial coefficient space, $J_N : \mathcal{P}_N \rightarrow E$ the full remainder map, and $M_N > 0$ its original source Gram. Keep

$$G = G_N = (J_N M_N^{-1} J_N^*)^{-1}. \quad (\text{FC1})$$

All adjoints denoted by $*$ are conjugate transposes in these specified coefficient frames. For a positive form T , write $B^{\dagger T} = T^{-1} B^* T$ and $\|B\|_T$ for its induced operator norm.

6.1.1 The actual finite sampling error

Fix the common source degree $D \geq N$, $a > 0$, integer $p \geq 1$, circle length $L > 0$ and integer cutoff $J \geq 1$. In the original diagonal log coordinates $r = \log x_k$, $z_i = \log x_i - \log x_k$, retain

$$(\mathcal{U}_k F)(r, z) = e^{(kr + \sum_i z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r), \quad \mathcal{K} = L^2(\mathbb{R}^{k-1}, dz). \quad (\text{FC2})$$

The Jacobian is $d^k x = e^{kr + \sum_i z_i} dr dz$, proving directly that this map preserves the original norm. Its derivative satisfies $\mathcal{U}_k D^{(k)} = (-\partial_r + k/2) \mathcal{U}_k$. Let Ψ_D be the image of the degree- D source columns. Define

$$\begin{aligned} M_{a,\pm,D} &= (e^{\pm ar} \Psi_D)^* (e^{\pm ar} \Psi_D), & H_{p,D} &= \int_{\mathbb{R}} (1+r^2) (\partial_r^p \Psi_D)^* (\partial_r^p \Psi_D) dr, \\ \kappa_{a,D} &= \max_{c \neq 0} \frac{c^* (M_{a,+D} + M_{a,-D}) c}{c^* M_D c}, & \lambda_{p,D} &= \max_{c \neq 0} \frac{c^* H_{p,D} c}{c^* M_D c}, \\ \eta_P &= \frac{\kappa_{a,D}}{e^{aL} - 1}, & \eta_T &= \frac{\lambda_{p,D}}{2p-1} \left(\frac{L}{2\pi J} \right)^{2p-1}, & \delta &= \eta_P + \eta_T. \end{aligned} \quad (\text{FC3})$$

These are finite observations of the original source; the weighted forms retain $x_k^{\pm 2a}$ and $1 + (\log x_k)^2$ in the original coordinates. For positive budgets $\varepsilon_P, \varepsilon_T$ with $\varepsilon_P + \varepsilon_T < 1$, choose the literal values of (P64):

$$L \geq a^{-1} \log(1 + \kappa_{a,D}/\varepsilon_P),$$

$$J \geq \max \left\{ 1, D, \frac{L}{2\pi} \left(\frac{\lambda_{p,D}}{(2p-1)\varepsilon_T} \right)^{1/(2p-1)} \right\}, \quad J \in \mathbb{Z}. \quad (\text{FC3a})$$

Substitution into (FC3) proves $\eta_P \leq \varepsilon_P$ and $\eta_T \leq \varepsilon_T$, including $\lambda_{p,D} = 0$. Thus $0 \leq \delta < 1$ has been obtained from actual finite quantities. No estimate uniform in h, k, D is inserted.

Use the plus Fourier transform $\hat{\psi}(u) = \int \psi(r) e^{iur} dr$. Let $M_N(L, J)$ be the full source Gram

$$M_N(L, J)_{ab} = \frac{1}{L} \sum_{|n| \leq J} \left\langle \widehat{\Psi_N e_a}(2\pi n/L), \widehat{\Psi_N e_b}(2\pi n/L) \right\rangle_{\mathcal{K}}. \quad (\text{FC4})$$

The $n = 0$ term is included with its original factor L^{-1} . For clarity, the estimates producing (FC3) can be checked without an inverse operation. Unfolding periodization gives $M_N(L) - M_N = \sum_{m \neq 0} \mathcal{C}_N(mL)$. Weighted Cauchy-Schwarz gives $|c^* \mathcal{C}_N(s) c| \leq e^{-a|s|} \sqrt{c^* M_{a,-,N} c} \sqrt{c^* M_{a,+,N} c}$. Summing both lattice directions bounds its absolute value by $c^*(M_{a,-,N} + M_{a,+,N})c / (e^{aL} - 1)$. Also integration by parts and $\int (1+r^2)^{-1} dr = \pi$ give $\|\widehat{\Psi_N c}(u)\|^2 \leq \pi |u|^{-2p} c^* H_{p,N} c$. Summing the two tails $|n| > J$, with $\sum_{n > J} n^{-2p} \leq J^{1-2p}/(2p-1)$, yields exactly the coefficient of $\lambda_{p,D}$ in (FC3). Restriction from the degree- D source preserves all these quadratic-form bounds. Therefore

$$(1 - \eta_P - \eta_T) M_N \preceq M_N(L, J) \preceq (1 + \eta_P) M_N. \quad (\text{FC5})$$

The asymmetry records that the omitted Fourier tail is positive. Put $\alpha = 1 - \eta_P - \eta_T > 0, \beta = 1 + \eta_P$, and retain

$$H = G_N(L, J) = (J_N M_N(L, J)^{-1} J_N^*)^{-1}. \quad (\text{FC6})$$

Completing the square proves that $v^* G v$ is the minimum of $c^* M_N c$ over $J_N c = v$, and the same statement holds for H . Taking these minima in (FC5) proves

$$\alpha G \preceq H \preceq \beta G, \quad (1 - \delta) G \preceq H \preceq (1 + \delta) G. \quad (\text{FC7})$$

The coefficient minimizers are $C_N = M_N^{-1} J_N^* G$ and $C_N(L, J) = M_N(L, J)^{-1} J_N^* H$. Their difference belongs to $\chi \mathcal{P}_{N-q}$. Fixed monic division writes its polynomial as $\chi Q = \sum_i h(s_i) Q_i(s)$. The original tensor primitive replaces F_h by ϕ_* in slot i , applies $Q_i(D_1, \dots, D_k)$, and has coefficient $(-1)^{i-1}$. The tensor differential contributes the same sign. Thus this change of representative has an explicit primitive with every original summand retained.

6.1.2 A dimension-free relative estimate for the residue coefficient

Let $I : F \hookrightarrow E$ be an actual nonzero proper A -invariant constituent and set $L_F = \ell I$. For every positive form T define

$$T_F = I^* T I, \quad P_{T,F} = I T_F^{-1} I^* T,$$

$$a_F(T) = \min_{x \in F} (e_0 - Ix)^* T (e_0 - Ix) = e_0^* T (1 - P_{T,F}) e_0, \quad (\text{FC8})$$

$$b_F(T) = L_F T_F^{-1} L_F^*, \quad \mathbf{c}_F(T) = a_F(T) b_F(T), \quad \sigma(T)^2 = (e_0^* T e_0) (\ell T^{-1} \ell^*).$$

Here $q > 1$. To verify strict positivity, $I(F)$ is an ideal of E because it is invariant under every polynomial in A . Its polynomial preimage is (g) with $\chi = gh$, and the original ideal basis is $[g], \dots, [S^{p_F-1} g]$, where $p_F = \deg h = \dim F$. Its last member is monic of degree $q - 1$, so $L_F \neq 0$; properness gives $e_0 \notin I(F)$. These statements prove $a_F(T), b_F(T) > 0$ without discarding repeated factors.

The same constrained-minimum argument as in (FC7) gives $\alpha a_F(G) \leq a_F(H) \leq \beta a_F(G)$. Compression and inverse order give $\beta^{-1} b_F(G) \leq b_F(H) \leq \alpha^{-1} b_F(G)$. Inverse order follows, for example, from the exact identity $w^* T^{-1} w = \max_x (2 \operatorname{Re} w^* x - x^* T x)$. Multiplication proves the full coefficient estimate

$$\boxed{\frac{\alpha}{\beta} \leq \frac{\mathfrak{c}_F(H)}{\mathfrak{c}_F(G)} \leq \frac{\beta}{\alpha}, \quad \left| \log \frac{\mathfrak{c}_F(H)}{\mathfrak{c}_F(G)} \right| \leq \log \frac{\beta}{\alpha}.} \quad (\text{FC9})$$

Applying the same two form bounds to e_0 and the inverse form to ℓ^* also proves

$$\frac{\alpha}{\beta} \sigma(G)^2 \leq \sigma(H)^2 \leq \frac{\beta}{\alpha} \sigma(G)^2. \quad (\text{FC10})$$

Every number is evaluated in its stated original or sampled metric. Neither (FC9) nor (FC10) has a dimension factor.

For the actual period matrix $\Pi(u, t)$, $u \neq 0$, put $T_{\text{per}}(t) = \Pi(t)^* \Pi(t)$ and let $m_H(t)$, $M_H(t)$ be the least and greatest eigenvalues of $H^{-1} T_{\text{per}}(t)$. Their positivity follows from the nonzero full period determinant. Applying the same calculation, now between H and T_{per} , gives

$$\frac{m_H(t)}{M_H(t)} \mathfrak{c}_F(H) \leq \mathfrak{c}_F(T_{\text{per}}(t)) \leq \frac{M_H(t)}{m_H(t)} \mathfrak{c}_F(H). \quad (\text{FC11})$$

The period equation $\Pi' = -\Pi(A + tR)/u$ implies, with $Y = \Pi I$ and $P = Y(Y^* Y)^{-1} Y^*$, $(1 - P)Y' = -t(1 - P)\Pi e_0 L_F/u$. Differentiating $\log \det(Y^* Y)$ gives $\partial_t \partial_t \log \det(Y^* Y) = \operatorname{Tr}((Y^* Y)^{-1} Y'^* (1 - P)Y')$. Consequently the precise connection to (FC9) is

$$\partial_t \partial_t \log \det(Y^* Y) = \frac{|t|^2}{|u|^2} \mathfrak{c}_F(T_{\text{per}}(t)). \quad (\text{FC12})$$

Equations (FC9)–(FC12) retain the actual comparison spectrum; they do not substitute the sampled coefficient for the period coefficient.

There is also a useful exact endpoint bound. For two admitted degrees $i \leq j$, the original representative spaces increase, so $G_i \succeq G_j > 0$. Let θ and ζ_+ be the least and greatest eigenvalues of the actual relative operator $G_i^{-1} G_j$. Then $\theta G_i \preceq G_j \preceq \zeta_+ G_i$ with $0 < \theta \leq \zeta_+ \leq 1$. Applying the proved minimum and inverse comparisons with these two factors gives the first inequality below. All eigenvalues lie in $(0, 1]$, so their product is at most their least member. This proves the remaining inequalities and retains the sharper endpoint coefficient from (RCX21):

$$\left| \log \frac{\mathfrak{c}_F(G_j)}{\mathfrak{c}_F(G_i)} \right| \leq \log \frac{\zeta_+}{\theta} \leq -\log \theta \leq \log \frac{\det G_i}{\det G_j}. \quad (\text{FC13})$$

The last inequality retains the correct direction. It does not infer an upper volume estimate from a scalar curvature observation. For the four original endpoints, retain the same degree $D = 2q + 2$ used to control their two generator raises. Its literal coefficient inclusion $I_{2q,D} : \mathcal{P}_{2q} \hookrightarrow \mathcal{P}_D$ restricts the two original forms to $M^{(0)} = M_{2q}$ and $M^{(1)} = M_{2q}(L, J)$. The complete signed determinant proof (PSA24a)–(PSA25) applies through these exact restrictions and the unchanged remainder maps. In particular it is an original/sampled comparison, with no replacement by the Gamma density. Write $b_1 \leq \dots \leq b_{2q+1}$ for the eigenvalues of $D_{L,J} = (M^{(0)})^{-1} M^{(1)}$. They lie in $[\alpha, \beta]$ by (FC5). The common-source projection formulas in (PSA24c) use $I_N B_N : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_{2q}$, where B_N itself has codomain $\mathcal{P}_N$; substituting $M_N = I_N^* M(x) I_N$ proves the equality of their relation Grams. The actual trace identity (PSA24d), full flag spectra, and retained cross-pairings prove

$$\begin{aligned} |\mathcal{B}(L, J) - \mathcal{B}| &\leq \mathcal{J}_{L,J}^{(5)} \leq \mathcal{J}_{L,J} \\ &\leq \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q-1) \log(\beta/\alpha). \end{aligned} \quad (\text{FC13a})$$

Here $\mathcal{J}_{L,J}$ is the generic two-bound integral in (PSA24e), and $\mathcal{J}_{L,J}^{(5)}$ is the actual-moment five-bound integral in (PSA24f). Its original measure and its positive atomic measure are explicitly identified there, so the RMT angle theorem applies through those same coefficient maps. The full nonlinear and signed interval (PSA25a) consequently propagates at these same four endpoints. Explicitly, if $\mathcal{B}_0 = \mathcal{B}$ is the original endpoint, define $a_q = 2q - 1$, $H_0 = 2 \operatorname{arcosh}(e^{\mathcal{B}_0/(2a_q)})$ and retain the five-entry radius $\mathcal{I}_{L,J}$ of (PSA24f). The constructed j_{ℓ_N}, e_{ℓ_N} in (PSA25ba)–(PSA25bc) use precisely $M_{2q}, M_{2q}(L, J)$, their original projectors and the same source inclusions; the finite-series index ℓ_N does not change the circle length. Thus the actual endpoint also obeys

$$\begin{aligned} \mathcal{B}(L, J) &\geq \max \left\{ 0, \mathcal{B}_0 - \mathcal{J}_{L,J}^{(5)}, 2a_q \log \cosh \frac{\max\{0, H_0 - \mathcal{I}_{L,J}\}}{2}, \mathcal{B}_0 + j_{\ell_N} - e_{\ell_N} \right\}, \\ \mathcal{B}(L, J) &\leq \min \left\{ \mathcal{B}_0 + \mathcal{J}_{L,J}^{(5)}, 2a_q \log \cosh \frac{H_0 + \mathcal{I}_{L,J}}{2}, \mathcal{B}_0 + j_{\ell_N} + e_{\ell_N} \right\}. \end{aligned} \quad (\text{FC13b})$$

Each bound is the proved same-source specialization of RMT22; taking the intersection preserves its exact endpoint. The formula and complete proof there retain all four signs; (FC13a) substitutes the same forms and bounds each relative ratio by β/α with total coefficient $1 + 2(q - 1) = 2q - 1$. The residue coefficient inequality (FC13) stays dimension-free: its operator is on the quotient E , while (FC13a) concerns the full source $\mathcal{P}_{2q}$. The exact map between them is the minimum section $C_N = M_N^{-1} J_N^* G_N$ from (FC7), satisfying $J_N C_N = I_E$ and $C_N^* M_N C_N = G_N$. It therefore retains the quotient norm and its original relation-valued sampling difference; no equality of these two different trace functionals is assumed.

6.1.3 Transporting the control form through the sampling error

The control also depends on the unchanged operator, so we retain its complete error term. For $t \in \mathbb{C}$, retaining that tensor degree k , set

$$\begin{aligned} A_t &= A + tR, \quad T_t = A_t - \frac{k}{2}I, \quad W_G(t) = A_t^* G + G A_t - kG, \quad X_G(t) = G^{-1} W_G(t), \\ W_H(t) &= A_t^* H + H A_t - kH, \quad X_H(t) = H^{-1} W_H(t), \quad B = G^{-1}(H - G). \end{aligned} \quad (\text{FC14})$$

The matrix identities in this subsection and in (FC24)–(FC29) also hold for any real scalar k when considered on the displayed finite coefficient space; the source realization retains its given integer tensor degree. Then $B^{\dagger G} = B$, $\|B\|_G \leq \delta$, and $H = G(I + B)$. No coordinate vector or inner product has been rescaled. Direct multiplication proves the exact identity

$$(I + B)X_H(t) = X_G(t) + T_t^{\dagger G} B + B T_t. \quad (\text{FC15})$$

Let $\epsilon_G(t) = \|X_G(t)\|_G$, $\epsilon_H(t) = \|X_H(t)\|_H$, and $\tau_G(t) = \|T_t\|_G$. The two X operators are self-adjoint in their respective metrics, so their norms are the suprema of their absolute quadratic forms divided by the respective squared norms. For every original v ,

$$|v^*(W_H(t) - W_G(t))v| = |\langle T_t v, Bv \rangle_G + \langle Bv, T_t v \rangle_G| \leq 2\delta\tau_G(t) v^* G v. \quad (\text{FC16})$$

Using (FC7) in both directions proves

$$\boxed{\epsilon_H(t) \leq \frac{\epsilon_G(t) + 2\delta\tau_G(t)}{\alpha}, \quad \epsilon_G(t) \leq \beta\epsilon_H(t) + 2\delta\tau_G(t).} \quad (\text{FC17})$$

Thus the control transfer retains the actual generator size as well as the relative metric error. Both quantities are finite observations of the original matrices. The exact LC additional radii $|t|\sigma(G)$ for $q > 1$, and $2|\operatorname{Re} t|$ for $q = 1$, remain available through (FC10).

For completeness, define the evaluated upper bound $e = \beta\epsilon_H(t) + 2\delta\tau_G(t)$ and $L_t = A_t^2 - kA_t$. Expansion gives $GL_t = W_G(t)T_t - T_t^* G T_t - k^2 G/4$. For each nonzero original vector v , put $z =$

$v^*GL_tv/(v^*Gv)$, $r = \|T_tv\|_G/\|v\|_G$ and $a + ib = \langle v, X_G(t)T_tv \rangle_G/(v^*Gv)$. Equation (FC17) gives $a^2 + b^2 \leq e^2r^2$, while the expansion gives $\operatorname{Re} z = a - r^2 - k^2/4$ and $\operatorname{Im} z = b$. The identities $er - r^2 = e^2/4 - (r - e/2)^2$ and

$$e^2 \left(\frac{e^2 - k^2}{4} - \operatorname{Re} z \right) - b^2 = (a - e^2/2)^2 + e^2r^2 - a^2 - b^2$$

prove the actual original-metric enclosure

$$\operatorname{Re} z \leq \frac{e^2 - k^2}{4}, \quad (\operatorname{Im} z)^2 \leq e^2 \left(\frac{e^2 - k^2}{4} - \operatorname{Re} z \right). \quad (\text{FC18})$$

No division by e occurs, so the zero-radius case is included.

6.1.4 The finite Fourier boundary and the quotient compensation

Let Q_J be the orthogonal Fourier projection to $|n| \leq J$, with the full $\mathcal{K}$ fibre at each retained mode. It commutes with $D_L = -\partial_r + k/2$. Use the actual least-norm representative for $M_N(L, J)$ and write

$$\begin{aligned} r^{L,J} &= \mathcal{V}_{h,k} C_N(L, J), \quad \mathcal{R} = Q_J \mathcal{P}_L \mathcal{U}_k r^{L,J}, \\ \mathcal{B} &= Q_J \mathcal{P}_L \mathcal{U}_k (D^{(k)} r^{L,J} - r^{L,J} A). \end{aligned} \quad (\text{FC19})$$

The explicit polynomial-division primitive following (FC7) applies to this relation as well. Direct intertwining gives $D_L \mathcal{R} - \mathcal{R} A = \mathcal{B}$ and $\mathcal{R}^* \mathcal{R} = H$. Since $D_L^* + D_L = k$ on these smooth finite-mode columns, multiplication and then one further application of D_L prove

$$\begin{aligned} W_H(0) &= -(\mathcal{R}^* \mathcal{B} + \mathcal{B}^* \mathcal{R}), \\ (D_L^2 - kD_L) \mathcal{R} - \mathcal{R} (A^2 - kA) &= (D_L - k) \mathcal{B} + \mathcal{B} A. \end{aligned} \quad (\text{FC20})$$

In particular (FC17) transports an actual boundary control computed from (FC20) to the original Gram. Finite mode count does not assign interval enclosures to its arithmetic coefficients; such enclosures have to be proved for any numerical application.

Finally retain the exact coefficient quotient $E/I(F)$ and a constant lift matrix J_F of its fixed basis. Then $C = [I, J_F]$ is invertible. For $T_{\text{per}} = \Pi^* \Pi$ the least-period-norm lift and its Gram are

$$\begin{aligned} Q(t) &= J_F - I(I^* T_{\text{per}} I)^{-1} I^* T_{\text{per}} J_F, \\ K(t) &= J_F^* T_{\text{per}} J_F - J_F^* T_{\text{per}} I(I^* T_{\text{per}} I)^{-1} I^* T_{\text{per}} J_F. \end{aligned} \quad (\text{FC21})$$

Completing the square in $\|\Pi(J_F v + Ix)\|^2$ proves uniqueness, positivity, $\pi Q = \pi J_F$ and $I^* T_{\text{per}} Q = 0$. The change $J_F \mapsto J_F + IB_0$ cancels exactly in Q and K ; a quotient basis change $J_F \mapsto J_F D_0$ gives $K \mapsto D_0^* K D_0$. Block elimination in $C^* T_{\text{per}} C$ gives

$$\det(I^* T_{\text{per}} I) \det K = |\det C|^2 |\det \Pi|^2. \quad (\text{FC22})$$

The complete period determinant is a nonzero constant times a holomorphic exponential in t at fixed u . Its retained squared constant is $(2\pi|u|)^q$; hence its log modulus squared has zero mixed curvature. Differentiating (FC22) and using (FC12) proves

$$\partial_{\bar{t}} \partial_t \log \det K(t) = -\frac{|t|^2}{|u|^2} \mathfrak{c}_F(T_{\text{per}}(t)). \quad (\text{FC23})$$

Both actual curvatures vanish at $t = 0$, and their nonzero leading coefficients have opposite signs. More precisely applying $\partial_{\bar{t}} \partial_t$ to (FC12) at zero gives $\partial_{\bar{t}}^2 \partial_t^2 \log \det(I^* T_{\text{per}} I)|_0 = \mathfrak{c}_F(T_{\text{per}}(0))/|u|^2$; the same derivative of $\log \det K$ is its negative. The respective $t^2 \bar{t}^2$ Taylor coefficients are these numbers divided by $2!2!$.

Apply the same Schur formula to the unchanged G and sampled H to obtain K_G, K_H . The fibre minima in the same quotient give $\alpha K_G \preceq K_H \preceq \beta K_G$. The literal comparison endomorphism $K_G^{-1}K(t)$ satisfies $K_G(K_G^{-1}K(t)) = K(t)$ and has log determinant curvature (FC23). This supplies the metric morphism on the original quotient, together with its finite sampling error and its exact curvature compensation. No coefficient class, generalized eigenspace or support label is removed by these identities.

6.1.5 The exact commutator allowance for the same two metrics

The coordinating Zeta task supplied the following sharper transfer, which we prove in full for the actual G, H, A_t already defined. It retains the specific metric discrepancy that interacts with the generator. All adjoints in this subsection are for G . Put $\mathcal{C} = G^{-1}H = I + B$, using the same B as in (FC14). This is a positive G -self-adjoint endomorphism with spectrum in $[\alpha, \beta]$. This symbol denotes the metric endomorphism; the constant quotient frame $C = [I, J_F]$ in (FC22) remains unchanged. Subtract $\mathcal{C}X_G(t)$ from (FC15), in the displayed order:

$$\boxed{X_H(t) - X_G(t) = \mathcal{C}^{-1}[T_t^{\dagger G}, B].} \quad (\text{FC24})$$

Indeed $X_G = T_t^{\dagger G} + T_t$, so the numerator of this difference is $T_t^{\dagger G}B - BT_t^{\dagger G}$.

Let $\mathcal{S} = \mathcal{C}^{1/2}$ be the unique positive G -self-adjoint square root. The finite spectral theorem constructs it by replacing each positive eigenvalue of the actual $\mathcal{C}$ by its positive square root. Uniqueness follows because a positive square root commutes with its square and, on each eigenspace of that square, positivity forces its eigenvalues to be that positive root. Thus no additional metric is selected. The exact map $\mathcal{S} : (E, H) \rightarrow (E, G)$ is an onto isometry, since $\mathcal{S}^*G\mathcal{S} = G\mathcal{S}^2 = G\mathcal{C} = H$. Its inverse is the actual $\mathcal{S}^{-1}$. In this comparison set

$$\begin{aligned} Y_H &= \mathcal{S}X_H\mathcal{S}^{-1} = \mathcal{S}^{-1}T_t^{\dagger G}\mathcal{S} + \mathcal{S}T_t\mathcal{S}^{-1}, \\ Z &= [\mathcal{S}, T_t]\mathcal{S}^{-1}, \quad Y_H - X_G = Z + Z^{\dagger G}. \end{aligned} \quad (\text{FC25})$$

The adjoint of Z is $\mathcal{S}^{-1}[T_t^{\dagger G}, \mathcal{S}]$; inserting it proves the last identity with both ordered terms present. The isometry gives $\|Y_H\|_G = \epsilon_H(t)$, so the reverse triangle inequality gives

$$|\epsilon_H(t) - \epsilon_G(t)| \leq \|Z + Z^{\dagger G}\|_G \leq 2\|[\mathcal{S}, T_t]\mathcal{S}^{-1}\|_G. \quad (\text{FC26})$$

The commutator $Y = [\mathcal{S}, T_t]$ satisfies the Sylvester equation $\mathcal{S}Y + Y\mathcal{S} = [\mathcal{C}, T_t] = [B, T_t]$. Its exact solution is

$$[\mathcal{S}, T_t] = \int_0^\infty e^{-r\mathcal{S}}[B, T_t]e^{-r\mathcal{S}}dr. \quad (\text{FC27})$$

To prove convergence, the original G -operator norm satisfies $\|e^{-r\mathcal{S}}\|_G \leq e^{-r\sqrt{\alpha}}$. Differentiating the integrand and integrating its two terms gives $\mathcal{S}Y + Y\mathcal{S} = [B, T_t]$, with value $[B, T_t]$ at zero and limit zero at infinity. For uniqueness, in the spectral decomposition of $\mathcal{S}$, each matrix block of a homogeneous solution is multiplied by the positive sum of its two eigenvalues; all such blocks must vanish. Therefore the integral equals the stated commutator, rather than a separate solution chosen for a bound. Its norm is at most $\|[B, T_t]\|_G/(2\sqrt{\alpha})$, while $\|\mathcal{S}^{-1}\|_G \leq \alpha^{-1/2}$. Substitution proves

$$\begin{aligned} \boxed{|\epsilon_H(t) - \epsilon_G(t)| \leq \|Z + Z^{\dagger G}\|_G \leq \alpha^{-1}\|[B, T_t]\|_G,} \\ [B, T_t] = [B, A] + t[B, R], \quad [B, R]v = (Be_0)\ell(v) - e_0\ell(Bv). \end{aligned} \quad (\text{FC28})$$

The scalar $k/2$ commutes exactly, and both residue terms remain. In particular the fully evaluated number

$$e_{\text{comm}}(t) = \epsilon_H(t) + \|Z + Z^{\dagger G}\|_G \leq \epsilon_H(t) + \alpha^{-1}\|[B, A] + t[B, R]\|_G \quad (\text{FC29})$$

is another valid original-metric control radius in (FC18). Taking the lesser of this number and the radius there remains valid because both have just been proved to bound $\epsilon_G(t)$. If $[B, T_t] = 0$, its G -adjoint gives $[T_t^{\dagger G}, B] = 0$, and uniqueness in (FC27) gives $[S, T_t] = 0$. Equations (FC24)–(FC26) then prove $X_H = X_G, Y_H = X_G$ and $\epsilon_H = \epsilon_G$ exactly.

These identities are covariant under every invertible coordinate map $U : E' \rightarrow E$. Indeed $G' = U^*GU, H' = U^*HU, T'_t = U^{-1}T_tU$ and $B' = U^{-1}BU$ imply $S' = U^{-1}SU$ by positive-square-root uniqueness in G' . Every commutator and integrand above transforms by this same similarity, and every norm is preserved by the stated coordinate map. Thus (FC28) is a comparison of the two original source observations with their full generator; its arithmetic commutator still requires evaluation for any numerical or uniform endpoint application.